Highlights

Lifted Gaussian Site Layers for In Context Sensor Fusion

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- Shared sensor noise can reverse the predictive role of an individual reading.
- Gaussian sites over a target and nuisance state support arbitrary sensor subsets.
- Support calibration initializes both cavity networks and transformer site layers.
- Completed experiments show synthetic gains and a useful real data tradeoff.

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ABSTRACT

A sensor deployment often has only a short calibration record and must continue to predict when some sensors become unavailable. The difficulty is that the sensors share disturbances, so a useful reading can change its role when another reading becomes available. We study probabilistic prediction from incomplete calibration data and arbitrary subsets of query sensors. We first characterize a limitation of Gaussian experts defined independently over the scalar target: when sensors share noise, experts that reproduce every singleton posterior cannot generally reproduce the posterior for a sensor pair. We then introduce support calibrated Gaussian site layers over the target and a small shared nuisance state. Under a linear Gaussian factor model, their product gives exact inference for each observed subset. A generic rank condition also identifies when the nuisance dimension is necessary for static sites. In the learned model, factor analysis of the support supplies an initial predictor, and a shared network refines each site using the posterior formed by the other sensors. The same construction provides a structured output layer for a transformer. Across the completed synthetic protocols, the compact cavity network improves likelihood under missing sensor counts absent from training and transfers from five sensors to sixteen. On the main panel it closes 58% of the reference gap, compared with 34% for the larger transformer trained on the same task distribution. The transformer site layer retains these synthetic benefits and improves the compact model on real sensor networks, while a conventional transformer remains stronger in several real conditions. The results identify a practical role for explicit shared noise structure in context based sensor calibration.

1. Introduction

A sensor calibration model is usually asked to do more than convert a voltage into a physical quantity. It must separate the quantity of interest from environmental influences, adapt to differences between instruments, and continue to work when some measurements disappear. These requirements are particularly visible in environmental monitoring, where inexpensive instruments are calibrated against a reference device for a limited period and then used without that reference [39, 2]. Calibration data are valuable, but they do not describe every combination of sensor availability that a deployment may encounter. A sensor may fail, be removed for servicing, or become unavailable at the same time as another sensor. A useful predictor therefore needs a distribution over the target for each available subset, together with a clear account of how the subset affects its uncertainty.

There are two different forms of adaptation in this problem. Across deployments, the response of a sensor can change in gain, offset, nonlinearity, and noise level. Within a deployment, the same collection of calibrated sensors can be observed in different subsets. Fitting a separate predictor for every subset uses the short calibration record repeatedly and does not scale well with the number of sensors. A single predictor with missing values filled by constants is easier to deploy, but it leaves the model to learn which parts of a filled input carry evidence and which parts merely encode absence. Neither choice directly explains how evidence from correlated measurements should be combined.

The correlation matters because several sensors can respond to a common disturbance. For example, an environmental variable can change several readings even when the target remains fixed. Once that disturbance has been marginalized, the readings are dependent conditional on the target. A product that assigns an independent likelihood factor to each sensor over the target alone then counts some shared evidence more than once. The resulting error is more specific than an overly narrow interval. Observing another sensor can change the optimal coefficient of an existing sensor, including its sign. A measurement that helps estimate the target in isolation can help estimate and subtract a disturbance when another measurement is present. Proposition 1 makes this obstruction precise for fixed scalar Gaussian experts.

We consider prediction from a calibration support set. The predictor receives labeled calibration rows, including their observation masks, and a query containing the available sensor readings. Its parameters are trained across deployments, while its adaptation to a new deployment comes from that support set. This is the setting of in context probabilistic prediction studied by conditional neural processes and prior fitted networks [8, 26, 14]. Here the task distribution supplies repeated opportunities to learn calibration, but the prediction must still respect the measurement structure of the particular deployment.

Our construction represents each observed sensor by a Gaussian site over an augmented variable containing both the target and a small shared nuisance vector. The sites are multiplied with a Gaussian prior, and the target marginal is obtained by a small matrix solve. Under a linear Gaussian factor model, this is ordinary conditional Gaussian inference written in a form that omits an unavailable sensor by leaving out its site. Proposition 2 establishes exactness under the stated model and mask assumptions. Proposition 3 explains why the nuisance dimensions are needed for generic shared noise when the sites are fixed. Together, these results give a statistical reason for retaining shared residual variation in the fusion state.

The initial sites come from a factor analysis model fitted to the incomplete support. We then study two ways to learn corrections. The compact lifted cavity network encodes each sensor's calibration scatter and repeatedly updates its site using the posterior implied by the other sensors. The lifted cavity transformer uses a cell token transformer to read the support and query, then predicts corrections to the same Gaussian sites. Its backbone can capture interactions across the complete observed context, while the final aggregation retains a Gaussian representation with positive variance. Both models begin at the fitted factor analysis predictor because their correction heads are initialized at zero.

This separation between evidence representation and learned correction is useful for interpretation. Fixed anchor sites provide a conditional under a fitted statistical model. Learned corrections adapt the response slope, offset, nuisance loading, and noise variance to the current context. The resulting distribution retains a transparent fusion calculation even when those parameters are predicted by a neural network. The small model and transformer therefore test the same site representation with two different ways of extracting information from calibration examples.

The empirical study uses synthetic sensor deployments with known generating distributions and two real sensor networks. It compares learned refinement with statistical calibration models and transformers trained on the same synthetic task family. The compact model benefits from its structure on the main synthetic tasks, including queries with deletion counts absent from training and panels wider than the training panels. The transformer site layer tests the same representation with a more flexible context encoder and extends the evidence to another real target. Response shifts and real data comparisons help establish where each encoder is most useful. The evaluation preserves the distinction between original checkpoints and a later repair for factor sign ambiguity, and reports every endpoint in the committed protocols. Calibration quality, missing sensor behavior, and computational cost are evaluated as parts of the same prediction problem.

2. Related work

2.1. Calibration and fusion with correlated measurements

Field calibration learns a relationship between sensor responses and a reference quantity under the conditions in which the instruments are used [2, 39]. Drift introduces an additional challenge because that relationship can change over time [34]. Our setting uses a short labeled calibration record for each deployment and asks for predictions after the reference target is hidden. It does not estimate a universal response curve for a named instrument. The support must instead identify the response and dependence structure needed for the current episode.

Correlated errors have also been studied in distributed estimation. Covariance intersection addresses fusion when correlations between estimates are unknown [17]. More recent neural approaches include IFNet and DIFNet for fusion with unknown correlations in measurement noise [35, 5]. DIFNet also considers changes in sensor nodes and network topology. KalmanNet combines learned components with filtering for partially known dynamics [29]. These approaches establish the importance of dependence aware fusion. The distinction here is the episodic calibration problem: sensor responses are inferred from labeled support rows, and the query is a contemporaneous subset of readings rather than a step in a specified state transition model.

2.2. Products of experts and learned approximate inference

A product of experts combines factors by multiplication [12]. In multimodal models, omitting a factor offers a convenient way to handle an absent modality [36, 22]. Multivariate latent products are already part of that literature, so lifting the prediction variable should not be read as introducing the general idea of a latent product. Our analysis concerns a narrower question: which variables must remain in a static Gaussian site representation if it is to reproduce sensor conditionals under shared noise? The scalar obstruction and the dimension result identify a limitation of target only factors under explicit assumptions. They do not exclude scalar sites that are recomputed from the full observation context.

Expectation propagation refines an approximate factor using a cavity distribution formed by removing the current factor [24]. Learned message operators can amortize those updates [11, 6, 15]. Our small model uses this organization: it removes one sensor's site, summarizes what the other sensors imply about target and nuisance, and predicts a replacement site. The unknown sensor response is inferred from calibration examples rather than supplied as a known likelihood. As a result, the model is inspired by expectation propagation but is trained by predictive likelihood, without an explicit moment matching constraint at every update.

Posterior linearization provides another connection. A nonlinear response can be approximated by a local linear Gaussian measurement under a current posterior approximation [3, 7]. We use the same interpretable quantities, namely a response offset, a target slope, nuisance loadings, and noise variance. The network learns those quantities from support and cavity statistics. This also follows the broader idea of algorithm unrolling, in which a finite sequence of inference operations is combined with trainable updates [10, 25]. The resulting guarantees are structural, such as positive posterior variance and insensitivity to hidden readings, rather than convergence guarantees for an inference algorithm.

2.3. Prediction with incomplete inputs

The observation process determines whether missingness can be ignored in inference [30, 21]. Supervised prediction with missing inputs also has statistical requirements that depend on the relationship between masks, inputs, and targets [16]. These considerations remain relevant even when a model is defined for every mask. A prediction rule that omits

absent likelihood factors is correct under its measurement model only when the mask supplies no further information needed for that conditional. Informative selection requires a model for selection or an explicitly different prediction objective.

NeuMiss derives a neural architecture from the Gaussian predictor with incomplete inputs [19]. It shares our interest in letting a statistical calculation guide network design. Its usual setting trains a predictor for a dataset with specified feature identities. Our model is trained across tasks and obtains each sensor's calibration from the current support. This distinction motivates shared sensor operations and a readout whose matrix dimension depends on the latent nuisance dimension rather than the number of available sensor columns.

2.4. Learning from a support set

Conditional and attentive neural processes learn predictive distributions conditioned on sets of examples [8, 18]. DeepSets and attention over sets provide reusable ways to encode unordered collections [37, 20]. The compact model uses a DeepSets encoder for the observed calibration pairs of one sensor. Its context exchange then occurs through Gaussian cavity statistics, which preserve an explicit target and nuisance interpretation.

Prior fitted networks train on tasks drawn from a prior so that a forward pass approximates inference for a new dataset [26]. TabPFN and its later tabular foundation model extend this approach to tabular prediction [13, 14]; the cited TabICL work studies classification [28]. Our transformer baseline adopts a cell token design inspired by TabPFN v2 and is trained on the sensor task distribution used in this study. It is not an evaluation of the published pretrained TabPFN model. The comparison between its direct Gaussian head and the lifted site head asks whether an explicit fusion calculation helps when the context encoder and training distribution are held fixed. Because the two heads also use different initializations, that comparison measures the complete readout choice rather than isolating a single source of improvement.

3. Problem setup

3.1. Deployment, support, and query

A deployment is represented by a task $\tau \sim p(\tau)$ with joint distribution $p_\tau(x, y)$. The sensor vector is $x \in \mathbb{R}^P$ and the target is $y \in \mathbb{R}$. The labeled support is

$$\mathcal{D} = \{(x_i, m_i, y_i)\}_{i=1}^n, \qquad m_i \in \{0,1\}^P, \tag{1}$$

where x_{ij} is available only if $m_{ij} = 1$. Every support row has an observed target. A query provides readings x_S for a subset $S \subseteq \{1, \dots, P\}$. Its mask is denoted m^q, so $S = \{j : m_j^q = 1\}$. The support mask and query mask have different roles: the former determines which calibration observations can be used, and the latter determines the evidence available for the current prediction.

The desired output is a predictive distribution for the hidden query target. Under a task prior and an ignorable observation process, the Bayesian target is

$$p(y \mid x_S, \mathcal{D}) = \int p_\tau(y \mid x_S)\, p(\tau \mid \mathcal{D}, x_S)\, d\tau. \tag{2}$$

The query can update beliefs about the task as well as about y. Thus integrating $p_\tau(y \mid x_S)$ against $p(\tau \mid \mathcal{D})$ without query reweighting is generally a different calculation. The learned models approximate Eq. (2) through training across tasks; they do not explicitly integrate a posterior over all calibration parameters at prediction time.

For the synthetic experiments, the oracle knows τ and evaluates $p_\tau(y \mid x_S)$. A separate Hamiltonian Monte Carlo reference integrates uncertain task parameters using the generating prior, after fixing response signs inferred from the support. Since those signs can be inferred incorrectly and their uncertainty is excluded, that numerical reference is not a certified Bayes optimal ceiling. Percentages described as closing a reference gap use its NLL difference from the reported FA Gaussian baseline. They should not be interpreted as a fraction of a proven information theoretic limit.

3.2. What missing sensor compatibility means

We require a predictor to be defined for any sensor subset and to be insensitive to the numerical contents of unobserved query entries. A stored placeholder in a masked cell must not alter the answer. We also consider whether the same learned parameters can be used for different numbers of sensors. These are architectural requirements. They are distinct from accurate uncertainty, from generalization to a new deletion count, and from consistency with a single joint distribution across all subsets.

The exact inference result in Proposition 2 assumes that the observation mask carries no additional information about the target or latent variables after conditioning on the variables in the measurement model. If a sensor disappears because its reading exceeds a threshold, for instance, its absence is itself evidence. Omitting its measurement factor does not encode that evidence. More generally, the target conditional is $p(y \mid x_S, \mathcal{D}, m^q)$ when selection is informative, and a selection model may be necessary. Our synthetic deletion mechanism is independent of readings and targets. Real data contain natural missingness as well as simulated deletion, so architectural compatibility does not by itself justify ignoring every aspect of the real observation process.

The distinction between construction and generalization is also relevant to the evaluation masks. Synthetic training queries omit at most one sensor. Synthetic evaluation additionally considers two or three missing sensors, as well as larger sensor panels. Those deletion counts are absent from the synthetic training queries. Real adaptation episodes retain their natural missing cells and remove at most one additional sensor; several sensors can therefore already be absent during that adaptation. We do not claim that every real evaluation deletion count is unseen during training.

3.3. Training objective and predictive representation

The learned predictor returns a Gaussian distribution,

$$q_\theta(y \mid x_S, \mathcal{D}) = \mathcal{N}(y; \mu_\theta, v_\theta),$$

with positive variance v_θ. Its episodic objective is

$$\mathcal{L}(\theta) = \mathbb{E}_{\tau,\mathcal{D},(x,y),m^q}\left[-\log q_\theta(y \mid x_S, \mathcal{D})\right]. \tag{3}$$

The expectation includes the training task prior and observation mechanism. NLL is a proper score for a predictive distribution [9]. Within the Gaussian family, minimizing its population expectation at a fixed observed context matches the conditional mean and variance, when these moments exist. This does not make the resulting Gaussian equal to a non Gaussian conditional density. In particular, multiple plausible target modes cannot be represented by a single Gaussian output.

The target is kept in its physical or episode specific target units at the final output. Sensor normalization is computed from observed support cells only. The models never use held out query targets to choose normalization constants, fit the anchor, or compute a site. The transformer also standardizes support targets internally, and its direct output is transformed back to the original target scale. The lifted readout instead works with a prior whose mean and variance are estimated from the support.

4. Why the shared state matters

The central question is what a sensor must contribute to a fusion rule. If the readings were independent after conditioning on the target, each sensor could contribute its own likelihood over that target. Shared disturbances change this picture. Another sensor can explain part of the first reading, and the direction of the remaining evidence can differ from the direction suggested by that reading alone. This section makes that issue precise before introducing the learned architecture.

4.1. A limitation of scalar experts

Consider the two sources of variation in

$$x_j = a_j y + d_j u + \sigma_j \epsilon_j, \tag{4}$$

where $y, u, \epsilon_1, \ldots, \epsilon_P$ are mutually independent standard normal variables, $\sigma_j > 0$, and $a_j \neq 0$. The coefficient a_j is the sensor's response to the quantity of interest. The coefficient d_j measures its response to a disturbance shared with the other sensors. A scalar product of experts uses

$$q(y \mid x_S) \;\propto\; p(y) \prod_{j \in S} t_j(y; x_j). \tag{5}$$

Each factor may depend on its own reading but is fixed with respect to the observed subset S. This restriction is useful because adding or removing a sensor requires only adding or removing a factor. It also creates an obstruction when that factor has to represent evidence whose meaning depends on another reading.

Proposition 1 (Scalar experts). *Let each t_j be Gaussian in y with positive precision, depend on x_j, and be independent of S. Suppose the product in Eq. (5) equals the Bayes posterior for every reading in every singleton subset. For a pair $\{j,k\}$, the product equals the Bayes posterior for every reading if and only if $d_j d_k = 0$. The product assigns x_j a posterior mean coefficient with the sign of a_j, whereas the exact coefficient can have the opposite sign.*

Matching all singleton posteriors fixes each factor, up to a multiplier independent of y, to its individual conditional likelihood. Multiplying these factors then treats the sensor errors as conditionally independent. In Eq. (4) their conditional covariance is $d_j d_k$, which the scalar factors cannot represent. Appendix A gives the full argument.

The sign change gives a concrete interpretation. Set $a = (1, 1)$, $d = (1, 0.5)$, and $\sigma = (0.05, 0.05)$. When the first sensor is available alone, its exact posterior mean coefficient is approximately 0.499. When both sensors are available, that coefficient is approximately -0.959. The second reading helps identify the shared disturbance, and subtracting that disturbance changes the first reading's contribution to the target estimate. Figure 1 illustrates this analytical example. The reversal concerns a conditional coefficient, rather than a change in the sensor's physical target response.

This proposition applies to static Gaussian experts over the scalar target. A factor that receives the full observed subset, a factor whose parameters depend on a joint cavity, or an expert over a richer latent state has additional information and is not covered by this result. The motivation for lifting is therefore constructive: retain the convenient product representation while giving it a state in which the common disturbance can be represented explicitly.

4.2. Exact fusion in a lifted state

Let $z = (y, h)$, where $h \in \mathbb{R}^K$ represents shared nuisance variation. Assume

$$\begin{aligned} y &\sim \mathcal{N}(\mu_y, v_y), \qquad h \sim \mathcal{N}(0, I_K), \\ x_j &= A_j y + D_j^\top h + c_j + e_j, \end{aligned} \tag{6}$$

with y and h independent, $v_y > 0$, and mutually independent $e_j \sim \mathcal{N}(0, \psi_j)$, $\psi_j > 0$, independent of z. Write $w_j = (A_j, D_j)$, $o_j = x_j - c_j$, and the prior natural parameters as (Λ_0, η_0). Sensor j's likelihood contributes

$$\Lambda_j = \frac{w_j w_j^\top}{\psi_j}, \qquad \eta_j = \frac{w_j o_j}{\psi_j}. \tag{7}$$

Its precision contribution has rank at most one and is positive semidefinite. The complete precision is positive definite because the prior is proper. The natural parameters combine by addition:

$$\begin{aligned} \Lambda_S &= \Lambda_0 + \sum_{j \in S} \Lambda_j, \\ \eta_S &= \eta_0 + \sum_{j \in S} \eta_j, \\ q(z \mid x_S) &= \mathcal{N}(\Lambda_S^{-1}\eta_S, \Lambda_S^{-1}). \end{aligned} \tag{8}$$

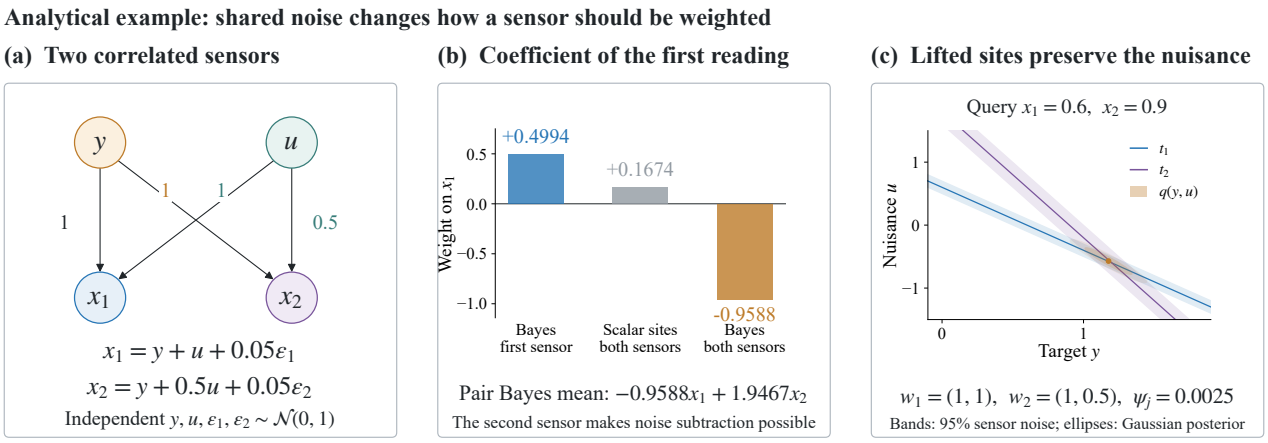


Figure 1: Analytical illustration of shared noise. Both sensors respond positively to the target, but they respond differently to the common disturbance. A single reading mixes these two sources. Their joint reading can separate them, allowing the first sensor's conditional coefficient to become negative. The coefficients are calculated from the stated Gaussian model and are not empirical performance measurements.

Proposition 2 (Exactness). *Under Eq. (6), the Gaussian in Eq. (8) is exactly $p(z \mid x_S)$ for every fixed subset S, including the empty subset. Its target marginal is the Bayes predictive distribution when the observation mask supplies no additional information about the target or nuisance state conditional on the observed readings.*

The result follows by expanding the quadratic form of each Gaussian likelihood. Missing sensors have no observed likelihood, so they contribute no site. The nuisance is then integrated out by taking the target marginal of the joint Gaussian. This is ordinary exact Gaussian conditioning expressed in a form that exposes the sensor contributions. The architectural choice is to preserve that form when the contributions are learned.

The qualification about masks is statistical. If a sensor is more likely to fail at high target values, the event of failure can itself contain evidence. In that case the relevant posterior also includes a likelihood for the mask. Omitting an unobserved value remains a well defined computation, but it does not account for information carried by the selection mechanism. Likewise, fitting the anchor from a short support set gives a plug in model. Exact inference in that fitted model does not integrate uncertainty over its estimated parameters.

4.3. When the nuisance dimension is necessary

Sufficiency follows directly from the factor model: a shared disturbance of dimension K can be represented with K nuisance coordinates. A converse requires specifying the family of admissible sites. If site parameters can depend arbitrarily on all readings and the selected subset, they can encode information beyond a static factor model. The following result concerns fixed linear Gaussian measurement sites.

Proposition 3 (Lifting dimension). *Let $x = ay + Du + e$, where $y \sim \mathcal{N}(0,1)$ and $u \sim \mathcal{N}(0, I_K)$ are independent, and the Gaussian errors e_j are mutually independent, independent of y, u, and have positive variance. Assume every $a_j \neq 0$. Write $L = [a, D]$, and suppose that after deleting any row, L contains two disjoint nonsingular $(1+K)$ by $(1+K)$ submatrices.*

Consider a proper Gaussian prior over $z = (y, h)$, $h \in \mathbb{R}^{K'}$, and static sites of the form in Eq. (7), with $o_j = x_j - c_j$, fixed w_j, c_j, and positive fixed ψ_j, all independent of readings and subsets. If their target marginal reproduces $p(y \mid x_S)$ for every reading and every subset with $|S| \leq 2$, then $K' \geq K$.

For almost every loading matrix L, the stated condition holds once $P \geq 2K + 3$. The proof first reconstructs the sensor covariance from the empty, singleton and pair conditionals. It then uses the disjoint full rank submatrices to identify the diagonal uniquenesses of the factor decomposition, following the classical identifiability argument of Anderson and Rubin [1]. A model with fewer latent coordinates cannot reproduce the resulting covariance. The full derivation is in Appendix A.

For the principal synthetic setting, $K = 1$ and $P = 5$. Under the generic condition, one shared nuisance coordinate is both sufficient and necessary within the static site family. This observation explains the choice of a small lifted state. It does not prescribe the latent dimension for arbitrary nonlinear sensor networks or replace model selection in a new application.

4.4. Properties of the learned construction

The compact cavity network described in Section 5 changes the site parameters using calibration information and cavity moments. Its outputs still define positive semidefinite precision contributions, but its learned predictive distribution need not be the exact posterior of the original physical model. Several useful construction properties remain.

Proposition 4 (Structural properties). *For every setting of the learned parameters, readings of missing query sensors do not affect the prediction. The same shared site network can be applied at any sensor count. Conditional on fully equivariant support statistics and consistently oriented anchor parameters, jointly permuting sensors and their calibration columns leaves the prediction unchanged. If the correction head is zero, every cavity round reproduces the Gaussian predictor of the fitted anchor.*

The first property follows because sites of missing sensors are masked out of the aggregate and therefore out of the cavities used to update observed sensors. Computing a local feature for a masked slot does not make its reading part of the fused evidence. The second follows from sharing the per sensor maps and using sums of natural parameters. The permutation property also requires the anchor representation to transform consistently. A factor loading and its sign reversed counterpart describe the same Gaussian model, but a generic learned map can respond differently to the two coordinates. This distinction is relevant to the original implementation and its replay audit, which is discussed in Appendix C.

The final property gives training a meaningful starting point. A zero correction head represents the support fitted closed form exactly, so the model class contains that predictor. Consequently the infimum of training risk over the class cannot exceed the anchor's training risk. This is a statement about representation, not a guarantee about optimization or performance on a new deployment. The empirical comparisons test whether learning makes useful use of the available corrections.

5. Lifted Gaussian site layers

Figure 2 follows a deployment from its incomplete calibration rows to the target distribution. The anchor gives the sites their initial statistical meaning. Learning changes those sites using the available context, while the Gaussian readout keeps the fusion calculation explicit. Figure 3 expands the update used by the compact network.

5.1. From correlated readings to a small Gaussian state

We retain the source of shared variation as part of the state used for fusion. Let

$$\begin{aligned} z = (y, h) \in \mathbb{R}^L, \qquad & L = 1 + K, \\ y \sim \mathcal{N}(\mu_y, v_y), \qquad & h \sim \mathcal{N}(0, I_K), \end{aligned} \tag{9}$$

with y and h independent and $v_y > 0$. The nuisance vector h is a statistical representation of shared residual variation; its coordinates need not identify a particular physical cause. The prior precision and natural mean parameter are

$$\begin{aligned} \Lambda_0 &= \operatorname{diag}(v_y^{-1}, I_K), \\ \eta_0 &= (\mu_y / v_y, 0, \dots, 0)^\top. \end{aligned} \tag{10}$$

For an observed sensor, a local linear Gaussian measurement with loading $w_j \in \mathbb{R}^L$, effective observation $o_j \in \mathbb{R}$, and variance $\psi_j > 0$ gives the site

$$\begin{aligned} t_j(z) &\propto \exp\left\{ -\tfrac{1}{2} z^\top \Lambda_j z + \eta_j^\top z \right\}, \\ \Lambda_j &= \frac{w_j w_j^\top}{\psi_j}, \qquad \eta_j = \frac{w_j o_j}{\psi_j}. \end{aligned} \tag{11}$$

The site precision is positive semidefinite and has rank at most one. A site need not be a normalized density over the whole state. Its interpretation is the likelihood of one scalar measurement, which constrains one direction in that state.

Aggregation uses only observed sensors:

$$\begin{aligned} \Lambda_S &= \Lambda_0 + \sum_{j=1}^{P} m_j^q \Lambda_j, \\ \eta_S &= \eta_0 + \sum_{j=1}^{P} m_j^q \eta_j, \\ V_S &= \Lambda_S^{-1}, \qquad m_S = V_S \eta_S. \end{aligned} \tag{12}$$

The predictive mean and variance are $m_{S,y}$ and $V_{S,yy}$. Since the prior precision is positive definite and the site contributions are positive semidefinite, the posterior precision is positive definite for every finite network output. If S is empty, Eq. (12) returns the fitted target prior. No artificial measurement is introduced for a missing sensor.

The calculation also shows how the nuisance affects target evidence. Partition the aggregate precision as $\Lambda_S = \left[\begin{smallmatrix} \alpha & b^\top \\ b & C \end{smallmatrix}\right]$ and its natural parameter as $\eta_S = (\eta_y, \eta_h)$. Integrating h gives

$$\begin{aligned} v_S &= (\alpha - b^\top C^{-1} b)^{-1}, \\ \mu_S &= v_S (\eta_y - b^\top C^{-1} \eta_h). \end{aligned} \tag{13}$$

The second term in each expression accounts for shared variation rather than counting each measurement as independent target evidence. Under the linear factor model of Proposition 2, this is the exact target conditional for every fixed observed subset. Proposition 3 explains the minimum nuisance dimension needed by fixed sites under its generic rank assumptions. These results establish the anchor calculation on which learned refinement builds.

5.2. Estimating the anchor from incomplete calibration rows

The anchor is fitted using the support alone. For sensor j, let $n_j = \sum_i m_{ij}$. We compute the observed mean $\bar{x}_j$ and observed standard deviation s_j, with a minimum scale of 0.1 in the implementation. Both support and query readings are expressed as $x_{ij}^s = (x_{ij} - \bar{x}_j)/s_j$ and $x_j^s = (x_j - \bar{x}_j)/s_j$. Missing support entries are stored as zero together with their masks. The zero is a storage convention: it is not treated as an observed calibration reading.

We fit a joint Gaussian for $r_i = (x_i^s, y_i)$ using expectation maximization with incomplete entries [4, 21]. In each iteration, let O_i denote the coordinates observed in row i, including the target coordinate. For the current mean μ and

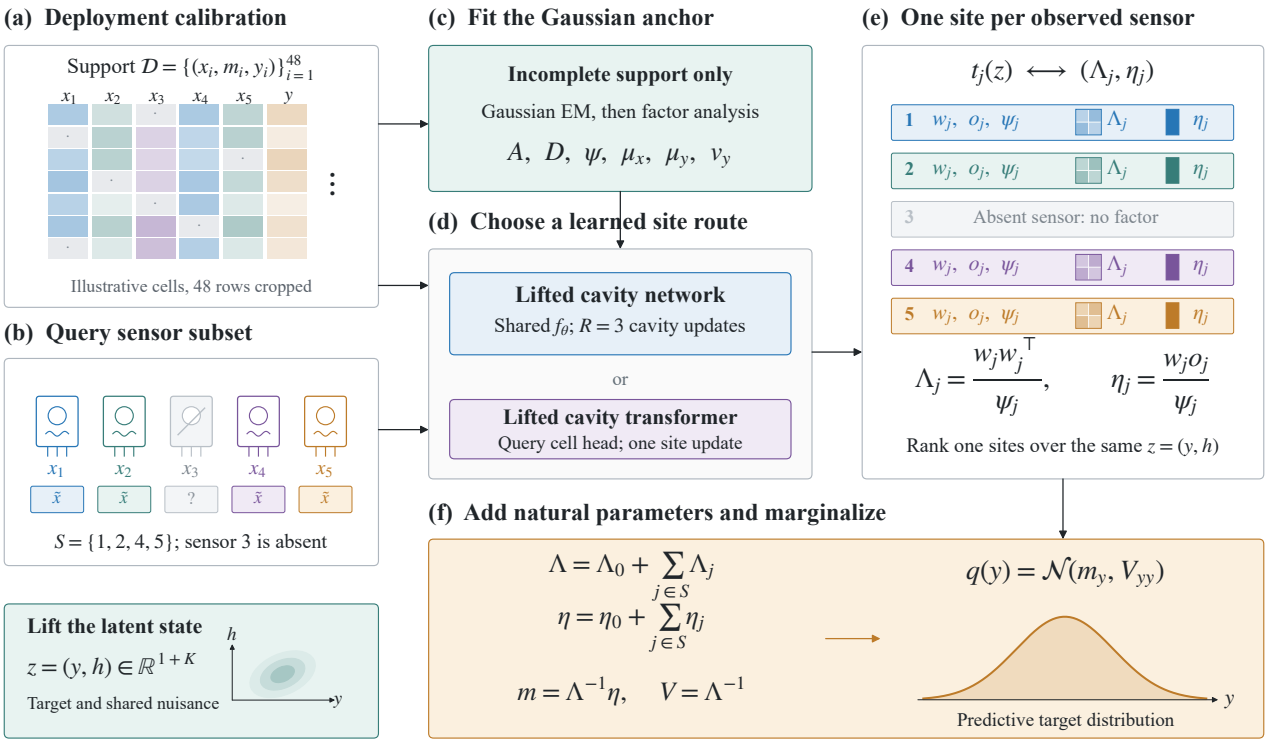


Figure 2: Architecture of the support calibrated Gaussian site predictor. The support matrix and query readings are illustrative; gray cells mark missing entries. A factor analysis anchor estimated from the incomplete support initializes the sites. Either a shared cavity network or a transformer supplies learned corrections. Only observed query sensors contribute natural parameters to the joint Gaussian over the target and nuisance state. Marginalization produces a predictive distribution for the target. The alternative routes are separate tested models.

covariance Σ, the conditional moments of the complete row are

$$\begin{aligned} G_i &= \Sigma_{:,O_i} \Sigma_{O_i,O_i}^{-1}, \\ \hat{r}_i &= \mu + G_i(r_{i,O_i} - \mu_{O_i}), \\ C_i &= \Sigma - G_i \Sigma_{O_i,:}. \end{aligned} \tag{14}$$

The covariance C_i is zero in fully observed directions. The next mean and covariance are

$$\begin{aligned} \mu^{\text{new}} &= \frac{1}{n} \sum_i \hat{r}_i, \\ d_i &= \hat{r}_i - \mu^{\text{new}}, \\ \Sigma^{\text{new}} &= \frac{1}{n} \sum_i \left[C_i + d_i d_i^\top \right] + 10^{-4} I. \end{aligned} \tag{15}$$

Including C_i is important: filling a missing entry only by its conditional mean would discard its remaining uncertainty. The implementation begins with observed means and a diagonal covariance, performs eight iterations for learned model anchors, and symmetrizes the covariance after each update. Anchor fitting uses double precision and is outside gradient propagation.

Write the fitted joint moments as $\mu_x, \mu_y, \Sigma_{xx}, \Sigma_{xy}$, and v_y. The supervised target loading and residual covariance are

$$A = \frac{\Sigma_{xy}}{v_y}, \qquad R = \Sigma_{xx} - AA^\top v_y. \tag{16}$$

We approximate R by $DD^\top + \text{diag}(\psi)$ using principal factor iterations, with $D \in \mathbb{R}^{P \times K}$. Starting from half the residual diagonal, floored at 10^{-3}, each iteration takes the leading K eigenvectors of $R - \text{diag}(\psi)$, scales them by the square roots of their nonnegative eigenvalues, and sets

$$\psi_j = \max\left\{ R_{jj} - \|D_j\|^2, 10^{-3} \right\}. \tag{17}$$

The batched anchor uses 40 such iterations. This is a concrete principal factor procedure; we do not claim that it finds the globally optimal factor analysis likelihood [31, 33]. The resulting fitted measurement model is

$$\begin{aligned} x_j^s &= \mu_{x,j} + A_j(y - \mu_y) + D_j^\top h + e_j, \\ e_j &\sim \mathcal{N}(0, \psi_j). \end{aligned} \tag{18}$$

Its sites use $w_j^{(0)} = (A_j, D_j)$ and $o_j^{(0)} = \tilde{x}_j + A_j \mu_y$, where $\tilde{x}_j = x_j^s - \mu_{x,j}$. Their target marginal is the FA Gaussian anchor. Exactness refers to this fitted model, not to the unknown deployment distribution. The separately

reported FA Gaussian baseline uses a longer 40 iteration Gaussian EM fit and a finite sample variance inflation; the learned anchors use eight EM iterations and no such inflation. They therefore share the same construction without being numerically identical baselines.

5.3. Encoding a sensor's calibration scatter

The anchor summarizes first and second moments. To expose departures from a linear response, each sensor also receives an encoding of its observed calibration scatter. Define

$$\begin{aligned} \widetilde{x}_{ij} &= x^s_{ij} - \mu_{x,j}, \\ r^{\mathrm{cal}}_{ij} &= \widetilde{x}_{ij} - A_j(y_i - \mu_y), \\ u_{ij} &= \left(\frac{y_i - \mu_y}{\sqrt{v_y}}, \quad \widetilde{x}_{ij}, \quad \frac{r^{\mathrm{cal}}_{ij}}{\sqrt{\psi_j}} \right). \end{aligned} \tag{19}$$

A shared DeepSets encoder [37] computes

$$e_j = \rho\left(\frac{\sum_i m_{ij}\phi(u_{ij})}{\max(n_j, 1)}, \frac{n_j}{n} \right) \in \mathbb{R}^{16}. \tag{20}$$

Here ϕ has widths 3, 32, 32 with tanh activations, and ρ maps the 32 pooled coordinates and observation rate to 16 coordinates with tanh. Every observed row contributes, while the mask excludes unavailable readings. Averaging makes the scatter encoding invariant to the ordering of the support rows. The observation rate records how much calibration evidence was available for that sensor, which the mean pooled features alone would not reveal.

5.4. Cavity conditioned site refinement

The compact model initializes the sites at the anchor and performs $R_{\mathrm{round}} = 3$ synchronous refinement rounds. In a round, sensor j removes its current natural parameters from the aggregate to form

$$\begin{aligned} V_{-j} &= (\Lambda_S - m^q_j \Lambda_j)^{-1}, \\ m_{-j} &= V_{-j}(\eta_S - m^q_j \eta_j). \end{aligned} \tag{21}$$

The cavity represents what the remaining observed sites imply about both target and nuisance. All replacements in a round are computed from the current sites before the next aggregation. A missing sensor's replacement is irrelevant because it contributes zero to that aggregation.

Let $\bar{y}_j = m_{-j,y}$. The anchor predicts the centered reading as

$$\begin{aligned} \widehat{x}_j &= A_j(\bar{y}_j - \mu_y) + D_j^\top m_{-j,h}, \\ r_j &= \widetilde{x}_j - \widehat{x}_j. \end{aligned} \tag{22}$$

The implementation measures its uncertainty by $v_j = w_j^\top V_{-j} w_j + \psi_j$, using the current site loading and variance. Thus the residual mean uses the anchor response, while its scale can reflect the current refined site. A shared network receives $\widetilde{x}_j$, e_j, A_j, D_j, $\log \psi^{(0)}_j$, and the observed sensor fraction $|S|/P$. Its cavity features are

$$\begin{aligned} &\left(\frac{\bar{y}_j - \mu_y}{\sqrt{v_y}}, \quad \log \frac{V_{-j,yy}}{v_y}, \quad m_{-j,h} \right), \\ &\left(V_{-j,yh}, \quad r_j, \quad \frac{r_j}{\sqrt{v_j}}, \quad \log v_j \right). \end{aligned} \tag{23}$$

For $K = 1$, these give seven cavity coordinates and 28 total inputs. The nuisance covariance block is not supplied in full. The feature vector is therefore a selected summary of the cavity rather than an unrestricted encoding of every covariance entry.

The site network has widths 28, 64, 64, 4 for $K = 1$, with tanh in its two hidden layers. It outputs response offset δ^{off}_j, target slope correction $\delta^{\mathrm{slope}}_j$, nuisance loading correction δ^D_j, and noise coordinate s^{noise}_j. In general there are $3 + K$ outputs. The replacement site is

$$\begin{aligned} a^{\mathrm{new}}_j &= A_j + \delta^{\mathrm{slope}}_j, \\ d^{\mathrm{new}}_j &= D_j + \delta^D_j, \\ g_j(\bar{y}_j) &= A_j(\bar{y}_j - \mu_y) + \delta^{\mathrm{off}}_j, \\ w^{\mathrm{new}}_j &= (a^{\mathrm{new}}_j, d^{\mathrm{new}}_j), \\ o^{\mathrm{new}}_j &= \widetilde{x}_j - g_j(\bar{y}_j) + a^{\mathrm{new}}_j \bar{y}_j, \\ \psi^{\mathrm{new}}_j &= \psi^{(0)}_j \exp\left[4 \tanh\left(s^{\mathrm{noise}}_j / 4 \right) \right]. \end{aligned} \tag{24}$$

This corresponds to the local measurement $\widetilde{x}_j \approx g_j(\bar{y}_j) + a^{\mathrm{new}}_j(y - \bar{y}_j) + (d^{\mathrm{new}}_j)^\top h + e_j$. The variance remains positive, with its ratio to the anchor variance between e^{-4} and e^4. Each round predicts a replacement relative to the anchor, rather than adding the correction repeatedly to the previous parameters. After the final round, Eq. (12) gives the Gaussian output. The encoder and site network together contain 8,004 parameters.

If the corrections came from statistical linear regression of a known response under the cavity, the construction would connect directly to posterior linearization and expectation propagation [3, 7]. Here the response is unknown and inferred from a short support. We train the finite sequence through its final predictive NLL. There is no separate loss enforcing expectation propagation moment matching, and no assertion that the rounds converge as their number increases.

5.5. What zero initialization and omission guarantee

The final affine layer of the correction network starts at zero. All four correction types therefore vanish initially. In Eq. (24), the effective observation reduces to

$$\begin{aligned} o^{\mathrm{new}}_j &= \widetilde{x}_j - A_j(\bar{y}_j - \mu_y) + A_j \bar{y}_j \\ &= \widetilde{x}_j + A_j \mu_y, \end{aligned} \tag{25}$$

and the site equals its anchor in every round. The model family thus contains the anchor exactly, providing a statistical

starting point for optimization. The guarantee concerns representation, while improvements after training are measured empirically.

Hidden query readings cannot influence an observed site's cavity because the aggregate includes only observed sites. The support encoder similarly excludes masked support entries. Consequently, changing a placeholder for an unavailable reading leaves the prediction unchanged. The shared encoder, shared site network, and sensor sums also permit the same parameters to operate at different widths. For fixed, consistently oriented anchor parameters, permuting sensor columns permutes the sites and leaves the readout unchanged.

The qualification about orientation matters. Factor analysis identifies $DD^\top$, while D is unchanged statistically by an orthogonal rotation of its columns. For $K = 1$ this includes replacing D by $-D$. The original network reads signed nuisance coordinates, so refitting the anchor after a sensor permutation can change its prediction through the eigensolver's orientation choice. Proposition 4 states the conditional invariance property. A separately versioned repair averages predictions under both signs for $K = 1$ and matches their first two moments. It addresses that sign ambiguity; arbitrary rotations for larger K and predictive evaluation of the repair remain separate questions. The efficacy tables concern the original checkpoints.

Likewise, factor omission must be distinguished from a projective joint model. Holding site parameters fixed, removing a sensor means deleting exactly its likelihood contribution. Once corrections depend on cavities or transformer context, the remaining site parameters can also change. The learned predictions are well defined for every subset but need not be the marginals of one shared joint density. The exactness and rank propositions apply to the fixed linear Gaussian construction within their assumptions, not to every context conditioned replacement.

5.6. The site layer with a transformer encoder

The lifted cavity transformer uses the same cell token backbone as the direct Gaussian transformer. Each support or query sensor cell is mapped from its scalar value to a 64 dimensional token; a masked cell uses a learned missing token instead. Every sensor column receives a random 16 dimensional feature vector, projected to token width and shared across rows within its task. The target column has a separate learned embedding. Support target values are standardized using the support mean and standard deviation, with a minimum scale of 0.1, while query targets receive a learned token.

Each of four transformer blocks first attends across the columns within a row, then across rows within a column. In the row attention operation, keys and values come only from support rows. Queries therefore do not attend to other queries, and adding another query does not supply evidence about the current query target. Each block uses four attention heads, normalization before each residual operation, and a feedforward network of hidden width 128 with GELU. The direct baseline reads the final query target token through a normalization and linear head with outputs (b_μ, b_ℓ). Using support target mean $\bar{y}_D$ and scale $s_{y,D}$, it returns

$$\begin{aligned} \mu &= \bar{y}_D + s_{y,D} b_\mu, \\ \ell &= 5 \tanh(b_\ell / 5) + 2 \log s_{y,D}. \end{aligned} \tag{26}$$

It contains 202,562 registered parameters.

For the lifted readout, each final observed query sensor token is mapped linearly to the $3 + K$ corrections in Eq. (24). The head is initialized at zero. The linearization point is the FA Gaussian posterior mean for that query, with its gradient detached, and there is one replacement and one aggregation. There are no three cavity refinement rounds in this model. The transformer may condition a sensor's correction on the complete support and observed query subset. Lifting constrains the output distribution and initialization, while preserving those context interactions.

For $K = 1$, the site head has 260 parameters and the implementation contains 202,822 registered parameters. This count includes the direct readout retained inside the instantiated backbone even though it is not used by the lifted forward calculation. Missing readings are replaced by the learned missing token before attention and their sites are omitted afterward. A missingness pattern can affect predictions, but the hidden numerical values cannot. Random column embeddings make the transformer symmetric in distribution over their draws; we do not assert exact permutation invariance for a fixed draw and reordered columns. Evaluation fixes the embedding seed. The comparison with the direct transformer changes both the output parameterization and its initialization, so their effects are not identified separately.

5.7. Optimization and deployment calculation

For predicted mean μ and log variance ℓ, the per query loss is

$$\mathcal{L}_q = \tfrac{1}{2} \left[\log(2\pi) + \ell + (y - \mu)^2 e^{-\ell} \right]. \tag{27}$$

The compact models train on a pool of 60,000 synthetic tasks, generated with pool seed 5150. A step samples 128 tasks and eight queries per task. The optimizer is AdamW with initial learning rate 1.8×10^{-3}, weight decay 10^{-4}, cosine decay to one tenth of the initial rate, gradient norm clipping at 5, and 20,000 steps. Reported compact runs use seeds 1, 2, and 3. Their main architecture uses $K = 1$ and three rounds. Support anchors can be precomputed for this pool because fitting does not depend on the learned parameters or query labels.

The transformer and lifted cavity transformer train on fresh tasks, with 16 tasks and 16 queries per step, learning rate 10^{-3}, 1,000 warmup steps, cosine decay, weight decay 10^{-4}, clipping at 1, and 30,000 steps. The lifted transformer fits its anchor for each newly generated task. Both use the same training query masks with at most one missing sensor. An additional direct transformer control sees uniformly sampled deletion counts from zero through four. No hyperparameter search was run for the compact architectures. Differences in task reuse and training budgets are

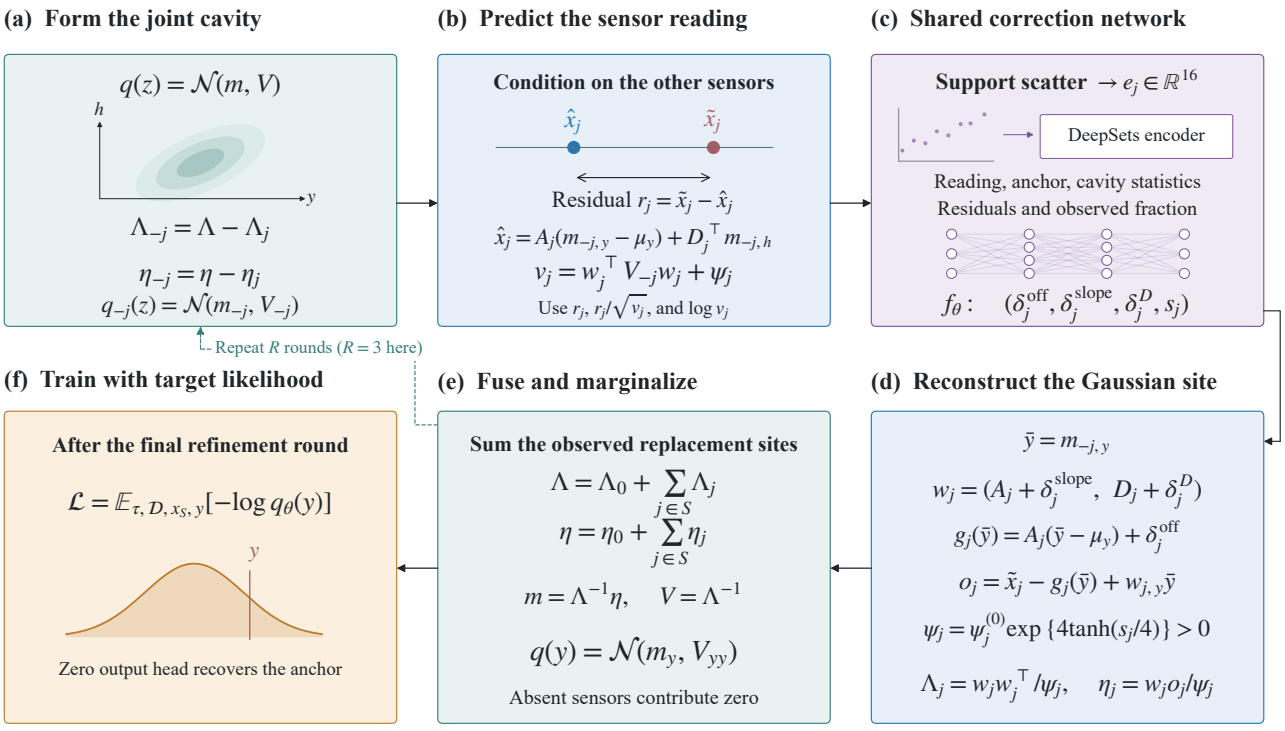


Figure 3: One round of cavity refinement. Removing sensor j's current site forms a joint target and nuisance posterior from the remaining observed sites. Its moments and the support fitted sensor response supply a predicted reading and residual features. The shared network returns corrections to the offset, slope, nuisance loading and log variance. The corrected site remains positive semidefinite in precision. Three synchronous rounds are used in the compact reported model.

retained in the comparison rather than described as identical optimization conditions across all model classes.

At deployment, observed support cells first determine normalization, the Gaussian EM fit, and the factor anchor. The compact model also computes the calibration scatter encodings once for that support. Each query then initializes its observed sites, performs the three refinement rounds, and returns the target marginal. The lifted transformer encodes support and query cells, predicts its one set of corrections, and performs the same final Gaussian aggregation. No optimizer update is needed for a new synthetic deployment. Real data adaptation, where used in the experiments, is a separate fine tuning stage.

Given the site parameters, aggregation and the dense latent solve cost $O(PL^2 + L^3)$ per query. Computing every cavity by dense inversion adds $O(PL^3)$ per compact refinement round. This cost excludes fitting the support covariance, principal factor iterations, calibration encoding, and transformer attention. For the main setting $L = 2$, the fusion state is small even when the sensor panel grows. Overall speed also depends on the context encoder and anchor fit, so the experiments report measured costs for the specified implementations.

6. Experimental design

The experiments ask whether retaining a shared nuisance variable helps a learned predictor combine incomplete sensor readings, and whether that benefit survives changes in sensor count, calibration size and data source. These are different questions. A model can transfer to a larger sensor set while still fitting real response functions poorly. We therefore retain the synthetic comparisons, the losses on real networks, the component ablations and the execution audits as separate pieces of evidence. All numerical results in this manuscript come from the existing research record. No model was fitted or selected when preparing this version.

6.1. Synthetic tasks and evaluation panels

For each task, the response of sensor j is generated as

$$x_j = a_j y + b_j + \alpha \sin(f_j y) + d_j u + \sigma_j \epsilon_j, \tag{28}$$

where the target y, shared nuisance u and sensor specific noise ϵ_j are independent standard normal variables. The loading a_j has a random sign and magnitude drawn uniformly from $[0.6, 1.8]$, $b_j \sim \mathcal{N}(0, 0.4^2)$, $f_j \sim \mathcal{U}[0.5, 2.5]$, $d_j \sim \mathcal{N}(0, 0.7^2)$ and $\sigma_j \sim \mathcal{U}[0.15, 0.8]$. Parameters are fixed within an episode and redrawn between tasks. The default nonlinearity is $\alpha = 0.4$. Sharing u makes the sensor errors dependent even when the target is known, while the

sinusoidal term introduces a controlled departure from the linear Gaussian assumptions of the anchor.

Each default episode contains 48 labeled support rows and 48 query rows. Every support sensor entry is removed independently with probability 0.2. Input centering and scaling use observed support entries, and query labels are used only for scoring. During synthetic training, a query has either no missing sensor or one missing sensor. Evaluation considers $k \in \{0, 1, 2, 3\}$ missing sensors. For a given k, every deletion pattern is scored when there are at most 20 patterns; otherwise, 20 patterns are sampled with a fixed seed. Query rows and masks are averaged within each task before computing the task level comparison. Consequently, hundreds of scores from the same task do not become hundreds of independent observations.

The main protocol v2 panel, F1, contains 256 new tasks with five sensors. Six additional panels contain 128 tasks each and change one factor at a time: eight sensors, sixteen sensors, $\alpha = 0.8$, $\alpha = 0$, 16 support rows or 96 support rows. The model trained on five sensors is used without adaptation on the wider panels. Protocol v3 adds G1 with 256 new tasks and five sensors, and G3 with 128 new tasks and sixteen sensors. These panels test the transformer readout hypothesis formed after the v2 results; reusing F1 for this hypothesis would have given only descriptive evidence.

6.2. Comparators and privileged references

The closed form comparators are fitted from the episode support only. FA Gaussian retains one residual factor, while EM Gaussian fits a joint Gaussian to the incomplete support using expectation maximization [4]. The reported closed form fits use 40 EM iterations. The neural anchor uses eight iterations, so initialization at the anchor does not make the untrained neural model identical to the stronger reported FA Gaussian baseline. On F1 with two missing sensors, the latter scores 0.177 and the untrained network scores 0.187. This distinction matters when attributing an improvement to learning.

NL FA fits cubic response curves in the standardized target, with an unpenalized intercept and slope, curvature penalty $\lambda = 100$, and a factor model for the residual covariance. Inference uses a 401 point grid. Curvature is clamped outside the support target range to obtain linear extrapolation. The polynomial degree and penalty were selected on development data; unrestricted selection by leaving out support rows produced unstable extrapolation. Bayesian linear regression uses evidence maximization and includes parameter uncertainty in its predictive variance [23]. Complete case ridge and the earlier ridge using mean imputation are retained to show the effect of strengthening the original control. When too few complete rows are available, the implementations use their recorded fallback rules, so the complete case labels describe the preferred fitting rule rather than an unconditional guarantee. The Gaussian process and nearest neighbour comparisons were added after the protocols and are descriptive.

The learned controls include a residual MLP with 15,490 parameters, the earlier scalar cavity architecture, a transformer with 202,562 parameters, and a transformer trained on every query mask. The residual MLP receives the same FA anchor and support encodings as the small lifted model, together with readings, masks and a support summary. Its sensor inputs are concatenated, so it is defined only for five sensors. The transformer follows the cell token design of TabPFN v2 [14], but is trained for this study from initialization. It is not the pretrained TabPFN v2 model. Four blocks alternate attention across columns and across support rows, using width 64 and four heads. Random feature embeddings allow it to operate at different sensor counts. Protocol v3 uses the same backbone with a lifted information site readout, giving 202,822 parameters.

We also evaluate two privileged references. The oracle knows the true generating parameters and evaluates the posterior numerically on a dense target grid. A second reference uses Hamiltonian Monte Carlo with the generator priors [27]. It integrates over response parameters and analytically marginalizes the shared nuisance, but fixes each response sign from its support correlation. Finite noisy support can identify that sign incorrectly, so this is a sign conditioned approximation rather than a certified Bayes optimum. It uses four chains, 50 retained draws per chain and query specific reweighting. Reported convergence diagnostics have $\widehat{R} < 1.1$ for at least 99.9% of identifiable parameters on the evaluated panels. Agreement with the oracle when true parameters are supplied checks the predictive calculation; it does not establish exact posterior exploration. Any percentage of a gap closed is relative to this particular HMC reference.

6.3. Training and real data episodes

The small learned models use a pool of 60,000 synthetic tasks generated with seed 5150. Training runs for 20,000 AdamW steps, with batches of 128 tasks and eight query rows per task, learning rate 1.8×10^{-3}, cosine decay to one tenth of that rate, weight decay 10^{-4} and gradient clipping at 5. The principal lifted model and stated controls use three training seeds. The transformers instead see newly generated tasks at every update and run for 30,000 steps with 16 tasks and 16 queries per task. They use learning rate 10^{-3}, 1,000 warmup steps, cosine decay and clipping at 1. The v3 comparison fixes the same recipe, seed and final checkpoint for both transformer variants. These comparisons match relevant inputs and masks, but do not match total task exposure or every training resource across all architectures. They therefore evaluate the recorded recipes rather than a universal parameter efficiency law.

Beijing multi site air quality [38] provides twelve monitoring stations. Each episode predicts a station's log pollutant concentration from contemporaneous readings at the other eleven stations. A week is eligible when the target station has at least 96 usable hours; 48 support hours and 48 distinct query hours are sampled within that week. Sensor gaps remain as observed in the raw data. Source adaptation uses March 2013 through December 2015, and evaluation

uses January 2016 through February 2017. Protocol v2 evaluates $PM_{2.5}$ on 718 station and week episodes. Protocol v3 evaluates previously unscored NO_2 and CO targets on 719 and 716 episodes. Each target uses its own eligible episodes. Conditions add either three or six sensor deletions to the natural query masks, while retaining an observed sensor whenever one exists. Support and query hours are sampled from the same week. This measures contemporaneous interpolation after local calibration, not prediction of a future time period from past hours alone.

The UCI Air Quality dataset [2] uses five metal oxide channels on one device to predict log CO or log NO_2 from reference analyzers. The loader removes the documented invalid readings and rows without the required sensor and target values before constructing episodes. Missing individual channels are then simulated, including 20% support dropout. The fitting period ends before October 1, 2004; the retained later test windows start on October 6 and comprise 23 weeks. CO was examined during development, and NO_2 was the v1 confirmation target. Their later reuse is secondary evidence on the same device, not replication across independent datasets.

Real data fine tuning uses one fixed recipe: 4,000 source episodes, 2,000 updates, learning rate 3×10^{-4} and random seed 11. Synthetic initialization, source only initialization and specified component controls are compared with the same adaptation rule. The gas sensor drift dataset [34] was inspected during development but excluded before building a test split: extreme readings made Gaussian mean NLL depend on a few episodes, with mean and median differing by more than 100 nats. This excluded benchmark remains a limitation of domain coverage.

6.4. Scores, intervals and declared decisions

Negative log predictive density in nats is the primary score. For Gaussian outputs with mean μ and variance v, it is $\frac{1}{2}\{\log(2\pi v) + (y-\mu)^2/v\}$. Lower values are better; negative NLL is possible for continuous densities. Secondary scores are coverage of the central 90% interval, continuous ranked probability score and RMSE [9]. Likelihood scores depend on the target scale and transformation, so values for different pollutants are not an absolute common accuracy scale.

Gains subtract the candidate NLL from its comparator NLL. A positive gain favors the candidate. Normal approximation intervals use the mean paired gain plus or minus 1.96 standard errors. Synthetic uncertainty is computed over tasks after averaging masks and the stated fitted seeds. Beijing differences are averaged across target stations within each week before computing the interval across weeks. This avoids counting stations sharing a week as independent, but does not establish independence of adjacent weeks. Averaging the NLL of fitted seeds is also distinct from scoring an ensemble density, and the intervals condition on those fits rather than incorporating all training randomness. The declared superiority gates generally require a gain of at least 0.01 nats and a positive lower bound. E7b instead requires a lower bound greater than -0.02 nats. These are nominal intervals for individual endpoints, with no multiplicity adjustment or declared overall superiority rule.

7. Results

7.1. Synthetic prediction and missing sensor patterns

Table 1 reports F1, including the privileged references and controls that did not perform well. Among methods fixed before this panel was scored, the small lifted cavity network has the lowest mean NLL at every tested deletion count. At two missing sensors, its NLL is 0.097, compared with 0.177 for FA Gaussian, 0.151 for NL FA, 0.131 for the transformer and 0.114 for the residual MLP. The paired gain over the transformer is +0.033 [+0.027, +0.040] nats. Both models were trained with at most one missing sensor, so this comparison includes query patterns absent from training.

Relative to the sign conditioned HMC reference, the lifted network closes 58% of the gap from FA Gaussian at two missing sensors. The residual MLP closes 46%, the transformer 34% and NL FA 19%. These ratios summarize this panel and reference; they do not estimate a fraction of all attainable performance. The oracle remains better, with NLL -0.026, demonstrating remaining headroom when the generating parameters are known.

The residual MLP is a useful close control because it receives the same anchor and support encodings. The lifted model improves on it by 0.030 [+0.022, +0.037] nats with every sensor observed, and 0.016 [+0.013, +0.020] with two missing. With three missing sensors the gain is only +0.002 [−0.001, +0.005]. This last interval includes zero. The evidence therefore supports a modest benefit over this MLP at the primary condition, rather than a clear separation at every deletion count.

Training the transformer on all masks yields NLL 0.132 at two missing sensors and closes 33% of the same reference gap. Expanded mask coverage does not remove its deficit under this recipe. That finding narrows the explanation beyond unfamiliar deletion patterns, but cannot by itself distinguish optimization, representation and differences in task exposure. EM Gaussian, Bayesian linear regression and both ridge controls are worse than the FA anchor on the primary condition. In particular, the ridge with mean imputation scores 0.399, showing why success against the original weaker control would have been insufficient evidence.

7.2. Changes in width, support size and nonlinearity

The wider panels provide the clearest transfer result in Table 2. The same lifted checkpoint runs at eight and sixteen sensors, although it was trained on five. At two missing sensors, its gains over FA Gaussian are +0.131 [+0.097, +0.165] and +0.148 [+0.115, +0.181] nats, respectively. The gain over the transformer at sixteen sensors is +0.144 [+0.127, +0.161]. The residual MLP cannot be evaluated at these widths. Its absence establishes an interface limitation

Table 1
Main synthetic panel (F1: 256 fresh tasks, five sensors). Mean test NLL by number of missing query sensors; learned models saw at most one missing sensor in training, so $k = 2, 3$ are unseen patterns. The lifted cavity and residual MLP rows average three training seeds (seed s.d. ≤ 0.002); the scalar cavity reference uses one seed, as do the transformers (seed 1 as declared in advance; Table 7 adds a second seed). Last column: share of the reference gap (FA Gaussian to sign conditioned HMC) closed at $k = 2$. Best nonprivileged entry per column in bold. †Post hoc, scored at $k = 2$ only. ‡Designed after protocol v2 (Section 7.5); descriptive on this panel and not bolded.

Method	k=0	k=1	k=2	k=3	Gap closed
Privileged references					
Oracle (true parameters)	−0.419	−0.250	−0.026	0.287	n/a
Sign conditioned HMC reference	−0.334	−0.174	0.039	0.337	n/a
Support only closed forms					
FA Gaussian	−0.152	−0.011	0.177	0.439	0%
EM Gaussian	−0.048	0.037	0.195	0.441	−13%
NL FA (cubic, tuned)	−0.173	−0.036	0.151	0.417	19%
Bayesian linear regression	0.111	0.091	0.214	0.450	−26%
Ridge (complete case)	0.026	0.078	0.216	0.451	−28%
Ridge (mean imputed)	0.198	0.280	0.399	0.582	−161%
GP, complete case†	n/a	n/a	0.416	n/a	−173%
Learned in context (trained on $k \leq 1$)					
Lifted cavity, 8k	**−0.265**	**−0.112**	**0.097**	**0.395**	58%
Transformer (TabPFN v2 style), 203k	−0.246	−0.084	0.131	0.424	34%
Residual MLP on FA anchor, 15k	−0.236	−0.087	0.114	0.397	46%
Scalar cavity network, 3k	0.055	0.178	0.336	0.555	−115%
Transformer, trained on all masks	−0.237	−0.076	0.132	0.411	33%
Lifted cavity transformer (v3)‡, 203k	−0.265	−0.112	0.095	0.387	59%

Table 2
Generalization beyond the training distribution (fresh panels of 128 tasks). Mean test NLL with two missing query sensors. Every learned model was trained on P=5 sensors, nonlinearity 0.4 and n=48 support rows; the residual MLP is defined only for five sensors. Best nonprivileged entry per column in bold; italic rows are controls or descriptive (the lifted cavity transformer was designed after protocol v2) and are not bolded.

Method	P=8	P=16	nonlin. 0.8	linear	n=16	n=96
Oracle	−0.598	−1.142	−0.086	−0.066	−0.060	−0.043
FA Gaussian	−0.296	−0.797	0.334	**0.009**	0.510	0.114
EM Gaussian	−0.160	2.977	0.311	0.034	1.140	0.112
NL FA	−0.327	−0.841	0.258	0.030	0.496	0.073
BLR	0.615	0.058	0.331	0.061	3.690	0.125
Lifted cavity	**−0.427**	**−0.945**	**0.191**	0.031	**0.363**	**0.052**
Transformer	−0.362	−0.801	**0.191**	0.085	0.523	0.081
Residual MLP	n/a	n/a	0.235	0.036	0.371	0.074
Transformer, all masks	−0.343	−0.790	0.190	0.080	0.483	0.080
Lifted cavity transformer (v3)	−0.423	−0.949	0.179	0.029	0.457	0.047

of this specific control, not a performance victory over every possible variable width network.

The unconstrained EM covariance fit deteriorates on sixteen sensors, where mean NLL reaches 2.977 compared with −0.797 for the factor anchor. This is consistent with unstable estimation from limited incomplete support. The lifted representation remains competitive in that setting. The transformer also accepts more sensors, but at sixteen its NLL of −0.801 is close to the FA anchor and materially worse than the lifted network.

With only sixteen support rows, the lifted network retains a +0.146 [+0.073, +0.219] nat gain over FA Gaussian, whereas the transformer has NLL 0.523 and the anchor 0.510. With 96 rows, the lifted model again has the lowest mean NLL among the methods fixed for v2. Stronger response nonlinearity produces a different result. At $\alpha = 0.8$ and two missing sensors, the lifted network and transformer have essentially identical means, with paired gain +0.000 [−0.011, +0.012]. With all sensors observed the transformer is better by 0.041 nats. Conversely, with truly linear sensors, the FA anchor is best: the learned model's gain over it

Table 3
Ablations on F1 (mean test NLL). Same training recipe for every learned row.

Variant	k=0	k=1	k=2	k=3
Lifted cavity, $K = 1$	−0.265	−0.112	0.097	0.395
Lifted cavity, $K = 2$	−0.267	−0.112	0.098	0.397
Lifted sites, no cavity input	−0.221	−0.074	0.123	0.399
Scalar sites ($K = 0$), same machinery	−0.070	0.061	0.232	0.473
Scalar cavity network (retrained)	0.055	0.178	0.336	0.555
FA Gaussian (untrained)	−0.152	−0.011	0.177	0.439

is −0.022 [−0.029, −0.015] nats at two missing sensors. Structure helps under several tested shifts, but learning a correction is not automatically beneficial when the anchor already matches the response family well.

7.3. Component ablations

Removing the nuisance dimension while retaining the rest of the small model changes NLL from 0.097 to 0.232 at two missing sensors. The paired difference is 0.134 [+0.120, +0.148] nats, and the scalar variant is worse than the untrained FA Gaussian baseline. Removing cavity inputs instead changes NLL to 0.123, a loss of 0.026 [+0.022, +0.029] nats. Adding a second nuisance dimension gives 0.098 and supplies no material benefit on a generator with one shared noise factor. Table 3 thus supports both lifting and cavity conditioning under the tested learning regime, with lifting accounting for the larger separation.

The scalar cavity architecture from the earlier study remains worse than the anchor after retraining with the larger shared task pool: its two deletion NLL is 0.336, or 0.159 nats worse than FA Gaussian. These empirical failures should not be read as a theorem about every scalar predictor. The structural argument concerns fixed local contributions; the context conditioned ablation has a broader input class and is evaluated here experimentally.

7.4. Real networks and the transformer advantage

On Beijing $PM_{2.5}$, Bayesian linear regression is the strongest reported closed form under natural missingness, with NLL 0.399. Fine tuning the lifted network reduces this to 0.205, a paired gain of +0.194 [+0.131, +0.257] nats. Its gain over BLR remains +0.137 [+0.104, +0.170] with six additional deletions. Synthetic pretraining helps relative to adaptation from initialization by 0.041 [+0.033, +0.050] nats under natural missingness.

The fine tuned transformer is nevertheless better than the lifted network: its NLL is 0.087, and the lifted model's gain is −0.118 [−0.145, −0.091]. This is the failed v2 endpoint E4b. The disadvantage persists after six extra deletions, with gain −0.069 [−0.080, −0.058]. Before adaptation, the ordering is reversed at that deletion condition: the lifted model is better by 0.127 [+0.057, +0.197]. Good transfer from the synthetic source therefore does not imply the best performance after target specific training.

On the Air Quality device, the fine tuned lifted network improves on FA Gaussian at two missing sensors by +0.107 [+0.015, +0.200] nats for CO and +0.181 [+0.076, +0.287] for NO_2. Its paired gains over the transformer are negative for both targets: −0.060 [−0.106, −0.014] and −0.044 [−0.074, −0.014]. Cavity conditioning also ceases to help on adapted CO, where the static lifted variant has lower NLL. These observations constrain any claim that cavity refinement itself reliably improves every real sensor task.

7.5. Lifted sites as the transformer readout

The v3 transformer readout isolates a narrower comparison: identical backbone, training recipe and seed, with a different probabilistic output layer. On the new G1 panel with two missing sensors, LCT gains +0.035 [+0.027, +0.042] nats over the ordinary transformer. On G3 with sixteen sensors, the gain is +0.150 [+0.133, +0.167]. Both declared endpoints pass. Relative to the small lifted network, LCT differs by +0.002 [−0.002, +0.006] on G1 and improves by +0.017 [+0.004, +0.029] on G3. The descriptive F1 result closes 59% of the HMC reference gap, but F1 was already used when this model was designed.

The recorded training curve reaches the ordinary transformer's final training NLL after 10,750 of 30,000 updates. Starting at a useful anchor plausibly contributes to this difference, since the ordinary head begins without that predictor. This is an optimization observation for the recorded curves, not a matched accuracy timing benchmark. On the previously used sixteen support row panel, LCT still scores 0.457, worse than the small model's 0.363. Giving the backbone a structured readout does not remove all of its sensitivity to limited calibration support.

On previously unscored Beijing NO_2, LCT reduces NLL from 0.282 for the small model to 0.220. The paired gain is +0.062 [+0.024, +0.100], so E7a passes. This recovers 75% of the mean gap between the small network and transformer for the protocol seed. Against the transformer itself, the gain is −0.021 [−0.093, +0.052]. Although the interval includes zero, its lower bound is below the fixed −0.02 margin. E7b therefore fails: absence of a clear superiority difference does not establish noninferiority. With six more stations removed, the paired gain is +0.005 [−0.027, +0.037], also too uncertain to establish a general ordering.

On CO, the descriptive comparison recovers 73% of the small model's gap under natural missingness, and the paired gain over the transformer is −0.035 [−0.057, −0.014] nats. Post hoc $PM_{2.5}$ and device analyses show partial recovery as well: 70% on Beijing $PM_{2.5}$ and 58% and 50% on device

Table 4
Real sensor networks: mean test NLL on later, held out periods. Beijing: 12 target stations × weekly episodes in 2016 to 2017 with natural missingness, plus extra dropped sensors. Air Quality: weekly episodes after October 2004, sensor dropout simulated. Fine tuning uses only earlier periods and one recipe for every model; lifted cavity and residual MLP rows average three seeds. Best entry per column in bold. †Post hoc. ‡Designed after protocol v2 (Section 7.5); post hoc here and not bolded.

Method	Beijing $PM_{2.5}$ (11 stations)			Air Quality CO		Air Quality NO_2	
	natural	+3	+6	k=0	k=2	k=0	k=2
EM Gaussian	1.032	0.742	0.569	0.074	0.085	−0.237	−0.175
FA Gaussian	0.677	0.615	0.562	0.052	0.116	−0.196	−0.155
NL FA	0.938	0.744	0.610	0.144	0.091	−0.090	−0.142
Bayesian linear regression	0.399	0.406	0.442	1.037	0.216	−0.055	−0.178
Ridge (complete case)	0.425	0.432	0.458	0.824	0.208	−0.200	−0.176
GP, complete case†	1.884	1.312	0.934	3.364	0.503	25.217	−0.198
k nearest neighbours†	0.508	0.514	0.540	0.571	0.353	−0.010	−0.063
Lifted cavity, before adaptation	0.826	0.661	0.561	0.044	0.129	−0.350	−0.233
Transformer, before adaptation	0.958	0.743	0.687	−0.017	0.081	−0.360	−0.231
Lifted cavity, fine tuned	0.205	0.237	0.305	−0.040	0.008	−0.412	−0.337
no synthetic pretraining	0.247	0.279	0.344	−0.001	−0.006	−0.419	−0.316
no cavity input	0.259	0.286	0.349	−0.086	−0.028	−0.423	−0.332
scalar sites ($K = 0$)	0.215	0.252	0.347	−0.026	−0.003	−0.359	−0.302
Transformer, fine tuned	**0.087**	**0.121**	**0.236**	**−0.163**	**−0.052**	**−0.485**	**−0.381**
Residual MLP, fine tuned	n/a	n/a	n/a	0.025	0.110	−0.246	−0.147
Lifted cavity transformer, fine tuned‡	0.122	0.160	0.259	−0.099	−0.027	−0.404	−0.359

Table 5
Protocol v3 (new panels and targets). Mean test NLL. G1: 256 fresh five sensor tasks by number of missing query sensors k; G3: 128 fresh 16 sensor tasks, $k = 2$. Beijing: log NO_2 and log CO at each station from the other 11 stations, 2016 to 2017, natural missingness and six further sensors removed; learned models fine tuned with the recipe of protocol v2. Best entry per column in bold.

	G1, P=5				G3	Beijing NO_2		Beijing CO	
Method	k=0	k=1	k=2	k=3	P=16	natural	+6	natural	+6
FA Gaussian	−0.189	−0.054	0.129	0.396	−0.806	0.684	0.506	0.384	0.334
Bayesian linear regression	0.013	0.022	0.153	0.401	0.063	0.275	0.365	0.170	0.248
Lifted cavity network, 8k	**−0.329**	**−0.172**	0.040	0.348	−0.934	0.282	0.334	0.149	0.229
Transformer, 203k	−0.315	−0.147	0.073	0.378	−0.800	**0.199**	0.296	**0.019**	**0.176**
Lifted cavity transformer, 203k	**−0.329**	**−0.172**	**0.039**	**0.340**	**−0.950**	0.220	**0.291**	0.054	0.187

CO and NO_2 at two deletions. With all device NO_2 channels present, LCT does not recover the gap. These results support a useful readout in some settings, while leaving real data superiority unresolved.

7.6. Calibration, training seeds and computational cost

Coverage helps distinguish likelihood gains from a predictor that is simply too confident. On F1 at two deletions, the lifted network has central 90% coverage 0.901, compared with 0.873 for FA Gaussian. Its CRPS and RMSE are 0.160 and 0.305, against 0.168 and 0.316 for that anchor. On Beijing, both fine tuned learned models have coverage 0.914, while FA Gaussian has 0.824. The transformer nevertheless has lower Beijing CRPS and RMSE than the lifted network. Near nominal marginal coverage therefore coexists with a meaningful difference in predictive sharpness and point accuracy; it does not establish conditional calibration.

The second transformer seeds in Table 7 were added post hoc. Synthetic gains retain their direction and positive intervals. The real NO_2 readout result is less stable: E7a changes from a gain of 0.062 for the protocol seed to 0.042 with an interval that includes zero for seed two. Gap recovery changes from 75% to 50%, or 62% after averaging the two seeds. The original E7b remains a failed noninferiority endpoint, and the additional seeds do not renew the confirmation claim.

Table 8 records training and inference on one CPU core. The small model uses 0.22 core hours and 7.2 ms per F1 task, compared with 3.11 core hours and 42.9 ms for the transformer. LCT uses 3.15 core hours and 55.7 ms. A task in this timing exercise contains 480 query evaluations. Neural anchor preprocessing is batched and excluded, while the closed form rows include their unbatched NumPy fitting costs. Thus these numbers describe the measured implementation, with an asymmetric preprocessing boundary; they are

Table 6
Calibration and point accuracy. Coverage of the central 90% predictive interval (target 0.90), continuous ranked probability score (CRPS, lower is better) and RMSE. F1 with two missing sensors; Beijing test with natural missingness (fine tuned learned models).

	F1, $k = 2$			Beijing, natural		
Method	Cov. 90	CRPS	RMSE	Cov. 90	CRPS	RMSE
FA Gaussian	0.873	0.168	0.316	0.824	0.189	0.379
EM Gaussian	0.860	0.170	0.318	0.807	0.207	0.423
BLR	0.866	0.174	0.327	0.861	0.182	0.374
Lifted cavity	0.901	0.160	0.305	0.914	0.171	0.357
Transformer	0.902	0.165	0.314	0.914	0.158	0.341
Residual MLP	0.902	0.162	0.309	n/a	n/a	n/a

Table 7
Second training seeds of the transformer and the lifted cavity transformer (post hoc). Endpoint comparisons recomputed with each seed and with both averaged; the lifted cavity network always averages its three seeds. Gains in nats, paired 95% intervals (computed across weeks on Beijing).

ID	Data	Comparison	Seed 1 (protocol)	Seed 2	Seeds averaged
E2	F1, $k = 2$	lifted vs transformer	+0.033 [+0.027, +0.040]	+0.037 [+0.030, +0.044]	+0.035 [+0.029, +0.042]
E3b	F3, $k = 2$	lifted vs transformer	+0.144 [+0.127, +0.161]	+0.155 [+0.139, +0.171]	+0.149 [+0.134, +0.165]
E5	G1, $k = 2$	LCT vs transformer	+0.035 [+0.027, +0.042]	+0.037 [+0.028, +0.046]	+0.036 [+0.028, +0.044]
E6	G3, $k = 2$	LCT vs transformer	+0.150 [+0.133, +0.167]	+0.161 [+0.142, +0.180]	+0.155 [+0.139, +0.172]
E4b	Beijing $PM_{2.5}$	lifted vs transformer (FT)	−0.118 [−0.145, −0.091]	−0.115 [−0.159, −0.070]	−0.116 [−0.150, −0.083]
E7a	Beijing NO_2	LCT vs lifted (FT)	+0.062 [+0.024, +0.100]	+0.042 [−0.023, +0.108]	+0.052 [+0.002, +0.103]
E7b	Beijing NO_2	LCT vs transformer (FT)	−0.021 [−0.093, +0.052]	−0.042 [−0.097, +0.012]	−0.032 [−0.093, +0.030]

not a complete deployment cost comparison. The larger v4 run is excluded from efficacy comparisons for the provenance reasons below.

8. Discussion and limitations

8.1. What the present evidence supports

The main finding is that a shared nuisance representation gives a small learned predictor a useful way to combine incomplete sensor readings. On the tested synthetic family, learning corrections to lifted Gaussian information sites improves prediction over the closed forms and ordinary transformer recipe. The width experiments show that the same sensor rule remains useful when additional sensors become available. The scalar ablation and static lifted control identify practical contributions from both the nuisance representation and the cavity inputs. These results build on products of experts, expectation propagation and learned messages [12, 24, 11, 6, 15], and connect those ideas to adaptation from a short calibration record.

The real benchmarks show the value of a flexible encoder. The small network improves on useful support fitted baselines, but the adapted transformer predicts better. Moving lifted sites to the transformer output layer recovers part of the deficit and preserves synthetic transfer. The failed noninferiority gate prevents treating that partial recovery as evidence that the readout has retained all of the transformer's real data flexibility. This distinction is substantive: the synthetic generator directly supplies the type of correlation represented by the nuisance factor, while real station and device responses can require richer interactions.

Post hoc diagnostics suggest underfitting by the small network. The transformer has lower NLL in 89% of Beijing $PM_{2.5}$ episodes, including differences of approximately 0.15 to 0.16 nats in each of the three easier quartiles of closed form difficulty. Its final training NLL is also lower by 0.19 nats. Increasing nuisance dimension from one to

Table 8
Compute. Training on one CPU core (core hours); inference (ms/task) is time on one thread per task for 480 queries (ten masks × 48 rows) on F1. Anchors are precomputed in batch and excluded; the closed form rows include their unbatched NumPy fits.

Model	Parameters	Training steps	Core hours	ms/task
Lifted cavity network	8,004	20,000	0.22	7.2
Residual MLP on FA anchor	15,490	20,000	0.20	1.2
Transformer (TabPFN v2 style)	202,562	30,000	3.11	42.9
Lifted cavity transformer	202,822	30,000	3.15	55.7
FA Gaussian (closed form)	n/a	n/a	n/a	38.3
NL FA (closed form)	n/a	n/a	n/a	34.4

Table 9
Post hoc fine tuning variants on Beijing $PM_{2.5}$ (test period of protocol v2; mean test NLL). Neither more nuisance dimensions nor longer fine tuning closes the gap to the fine tuned transformer.

Fine tuned model	natural	+3	+6
Lifted cavity network, standard recipe (seed 1)	0.207	0.237	0.304
with $K = 2$	0.205	0.239	0.300
with 5× fine tuning steps	0.169	0.201	0.276
Transformer, standard recipe	0.087	0.121	0.236

two changes the exploratory natural mask score from 0.207 to 0.205; increasing adaptation steps fivefold improves it to 0.169, still above the transformer's 0.087. These results make a limitation of the local correction inputs plausible, but they do not prove the mechanism. The readout experiment tests one response to that hypothesis, and its real outcome remains seed sensitive.

The results suggest useful next comparisons. Set processing supports interactions among variable numbers of inputs [37, 20]; matching a flexible set readout and a direct covariance or precision readout would isolate the practical value of the information site construction. The 203k transformer establishes a controlled study comparison. Evaluating pretrained tabular systems [14, 28] would address a complementary question about current predictive performance. Pretrained TabPFN belongs to the unverified v4 study, while TabICL remains unevaluated.

8.2. Provenance and interpretation

The experimental record includes advance protocols, retained scores and inference replay. The five original confirmation decisions reproduce in aggregate, although factor orientation prevents exact replay of every historical score cell. The larger three seed v4 continuation also completed and its artifacts were collected. Independent verification stopped before endpoint arithmetic because required checkpoint identities had not been bound before scoring, so its five endpoints remain unverified and withheld. The quantitative findings here therefore come from v1 to v3. Appendix C provides the full protocol outcomes, orientation diagnostics and v4 audit, including the prospective engineering repairs and their distinct version boundaries.

8.3. Statistical and domain limits

The synthetic family deliberately has Gaussian targets, one shared nuisance, independent idiosyncratic noise and sinusoidal response deviations. The exactness arguments concern static linear Gaussian factors, while the learned nonlinear corrections are assessed empirically. Gaussian predictions cannot represent genuinely multimodal target posteriors. Neither the ablations nor the tests establish robustness to arbitrary correlated errors, changing latent rank, informative outages or a causal missingness mechanism [30, 21].

Natural missingness accounts for only 1.6% of Beijing query readings in the $PM_{2.5}$ panel, and most stress conditions remove additional sensors artificially. The device benchmark begins with complete required measurements and then simulates dropout. Both settings are useful controlled interpolation tasks, but they give limited evidence about naturally failing networks. Pollutants from the same stations and targets from the same device are related observations, not independent domain replications. The excluded drift benchmark further narrows the evidence. No clinical, exposure decision or future forecasting benefit has been evaluated.

Paired intervals quantify task or week variation conditional on the tested fits. They do not include all uncertainty from architecture development, hyperparameter choices, training randomness, temporal dependence or repeated endpoints. The second seed analysis illustrates why a narrow task interval cannot replace independent fitted seeds, especially for the real readout result. Marginal 90% coverage is also weaker than calibration within each station, concentration range or missingness condition. Before operational use, performance and interval behavior would need validation in the intended deployment. These limits leave a useful research finding about structured readouts, while keeping broader effectiveness claims open.

9. Conclusion

Lifted Gaussian information sites provide a concrete way to retain shared sensor noise when predictions must use changing subsets of observations. Their exact interpretation holds under the stated linear Gaussian assumptions. In the recorded synthetic experiments, a small learned cavity network improves on the tested support fitted methods and ordinary transformer recipe, including transfer from five to sixteen sensors. A lifted transformer readout preserves that benefit and improves on its matched ordinary head on new synthetic panels.

On real networks, combining a flexible transformer with the site readout recovers part of the gap left by the small cavity network. The conventional transformer retains an advantage in several conditions, and the declared noninferiority margin is not established. Together, the theory and experiments identify a useful design principle: retain common disturbances in the fusion state, and learn from calibration data how each observed sensor should modify that state.

Data and code availability

The implementation, historical experiment protocols and score exports are available in the MIRA repository at `https:`

//github.com/saptarshihalder/MIRA. The quantitative tables in this manuscript retain the numerical values of the existing lifted cavity paper. The accompanying source package includes the bibliography, the unchanged Elsevier class and style files, the figure sources and a record connecting the manuscript to the source revision. Public availability of code and score summaries does not by itself establish reproduction of every trained checkpoint. The distinctions between aggregate result checks, prediction replay and the later provenance audit are documented in Appendix C.

Declaration of competing interest

The authors will supply their affiliations and declarations before submission. Author information in this draft is anonymous.

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A. Proofs

Proposition 1. The individual conditional is

$$p(x_j \mid y) = \mathcal{N}(x_j; a_j y, c_j), \qquad c_j = d_j^2 + \sigma_j^2.$$

If $q(y \mid x_j) = p(y \mid x_j)$ for all x_j, then $p(y)\, t_j(y; x_j) \propto p(y)\, p(x_j \mid y)$ as functions of y. Hence

$$t_j(y; x_j) = \kappa_j(x_j)\, p(x_j \mid y).$$

This factor is Gaussian in y with precision $a_j^2/c_j > 0$. For a pair, q is therefore the naive Bayes posterior $p(y)\, p(x_j \mid y)\, p(x_k \mid y)$. Its posterior mean is linear in (x_j, x_k) with coefficients

$$(a_j/c_j,\ a_k/c_k)\,/\,\pi_q, \qquad \pi_q = 1 + a_j^2/c_j + a_k^2/c_k,$$

so the coefficient of x_j has the sign of a_j.

The Bayes posterior uses $C = \mathrm{Cov}(x_j, x_k \mid y)$, with off diagonal entry $d_j d_k$. Its mean coefficients are $(C^{-1}a)^\top/\pi$, with $\pi = 1 + a^\top C^{-1} a$. Both posteriors are Gaussian, so they coincide for all (x_j, x_k) if and only if $C^{-1}a = \mathrm{diag}(C)^{-1}a$. Equivalently $a = C\,\mathrm{diag}(C)^{-1}a$, whose off diagonal components are $d_j d_k\, a_k/c_k = 0$ and $d_j d_k\, a_j/c_j = 0$. With $a_j a_k \neq 0$ this holds if and only if $d_j d_k = 0$.

For the sign flip, take $a = (1, 1)$, $d = (1, 0.5)$ and $\sigma = (0.05, 0.05)$. Then the Bayes coefficient of x_1 is $+0.499$ alone and -0.959 in the pair. □

Proposition 2. Given $z = (y, h)$, the readings are independent: $x_j \mid z \sim \mathcal{N}(w_j^\top z + c_j, \psi_j)$. Hence

$$p(z \mid x_S) \propto p(z) \prod_{j \in S} \mathcal{N}(x_j; w_j^\top z + c_j, \psi_j).$$

In natural parameters each factor contributes $w_j w_j^\top/\psi_j$ to the precision and $w_j(x_j - c_j)/\psi_j$ to the shift, which is (7) with $o_j = x_j - c_j$. The prior is Gaussian, so the product is the Gaussian (8). This holds for every S because the factors of unobserved sensors are absent from both sides. Marginalizing h gives $p(y \mid x_S)$. □

Proposition 3. Step 1: the product is a latent factor model. By the computation in the proof of Proposition 2 read backwards, a product of sites (7) with $o_j = x_j - c_j$ is the exact posterior over z of the model M_q: $z \sim \mathcal{N}(\mu_z, \Sigma_z)$ and, independently given z, $x_j \sim \mathcal{N}(w_j^\top z + c_j, \psi_j)$. Its y-marginal is M_q's conditional $p_q(y \mid x_S)$, and M_q's sensor covariance is $\Sigma_q = W \Sigma_z W^\top + \mathrm{diag}(\psi)$ with $W \in \mathbb{R}^{P \times (1+K')}$, so $\Sigma_q - \mathrm{diag}(\psi)$ has rank at most $1 + K'$.

Step 2: conditionals on $|S| \leq 2$ identify the covariance. Write $V = \mathrm{Var}(y)$, $c = \mathrm{Cov}(x, y)$ and $\Sigma = \mathrm{Cov}(x)$. The case $S = \emptyset$ gives V. For $S = \{j\}$ the conditional has slope $\beta_j = c_j/\Sigma_{jj}$ and variance $r_j = V - c_j\beta_j$, and $\beta_j \neq 0$ because $c_j = a_j \neq 0$. Hence $c_j = (V - r_j)/\beta_j$ and $\Sigma_{jj} = c_j/\beta_j$. For $S = \{j, k\}$ the slope vector $b \neq 0$ solves $\Sigma_{\{jk\}} b = (c_j, c_k)$, and whichever entry of b is nonzero determines Σ_{jk} from one row. Since M_q has the same conditionals, the same formulas apply to it, so $\Sigma_q = \Sigma = LL^\top + \mathrm{diag}(\sigma^2)$.

Step 3: fewer factors cannot reproduce Σ. Suppose toward contradiction $K' < K$. Let $r = 1 + K$, so $LL^\top$ has rank r. Suppose $\Sigma = G + \Psi'$ with Ψ' diagonal and $\mathrm{rank}(G) \leq r$. Fix a row i and two disjoint row sets I_1, I_2 not containing i with L_{I_1}, L_{I_2} nonsingular. In the $(r+1) \times (r+1)$ submatrix of Σ with rows $\{i\} \cup I_1$ and columns $\{i\} \cup I_2$, every entry except (i, i) lies off the diagonal, so it is shared by G and $LL^\top$. Both G and $LL^\top$ have rank at most r, so the corresponding minors vanish. Expanding along entry (i, i), its cofactor is $\det(L_{I_1} L_{I_2}^\top) \neq 0$, so $G_{ii} = (LL^\top)_{ii}$. Hence $G = LL^\top$, which has rank r [1]. Applying this to $G = W \Sigma_z W^\top$ from Step 1 gives $1 + K' \geq r = 1 + K$.

For almost every L all $r \times r$ submatrices are nonsingular, and $P \geq 2r + 1 = 2K + 3$ leaves two disjoint sets of r rows after deleting any row. □

Proposition 4.

(i) Sites of missing sensors are multiplied by the mask before entering (8), and the network's per sensor inputs never include another sensor's raw reading, only cavity statistics built from observed sites.

(ii) Every operation is applied per sensor with shared weights. Conditional on a fully equivariant, consistently oriented anchor representation, its fitted sensor parameters permute with the sensor coordinates, and sums over sensors are permutation invariant. An unconstrained eigensolver can change factor signs or rotations when the anchor is refitted; the original learned network is not invariant to these changes. No input encodes a sensor's index or the total number of sensors, except the fraction of observed sensors, which is permutation invariant.

(iii) With a zero last layer, $\delta^{\mathrm{off}} = \delta^{\mathrm{slope}} = \delta^D = s = 0$. Then $w_j = (A_j, D_j)$, ψ_j is unchanged and $o_j = \tilde{x}_j - A_j(\bar{y} - \mu_y) + A_j \bar{y} = \tilde{x}_j + A_j \mu_y$. These are the initialization sites, so every round reproduces them and the output is the anchor's FA Gaussian closed form. The zero point lies in the parameter set, so the infimum of the training risk is at most that closed form's. □

B. Reference inference and implementation details

B.1. The oracle and the fitted Gaussian reference

The oracle uses the actual parameters of the synthetic deployment. It conditions on the observed sensor subset under that deployment and therefore measures performance with task uncertainty removed. Its role is diagnostic. It shows whether an inference mechanism loses information even when its sensor model is known, and it supplies a lower risk reference under the specified generator.

FA Gaussian instead estimates its parameters from the incomplete support. The network anchor uses eight EM iterations, whereas the reported closed form reference uses forty. Both fit the joint Gaussian before extracting the factor structure. The stronger closed form reference is not numerically identical to the initial network in every episode. This distinction matters when interpreting the gain from learning: the final network is compared against the specified reference, rather than against a deliberately identical numerical starting point.

The reported EM Gaussian, FA Gaussian and ridge references inflate predictive variance by $n/(n-|S|-1)$. This is part of the reference implementation, not a consequence of the exact site identity. The learned models predict their own corrected variances. Comparisons therefore evaluate the complete specified predictors, including their treatment of finite support uncertainty.

B.2. The sign conditioned sampling reference

The sampling reference approximates inference over the synthetic task parameters from the support, using the generator prior [27]. There are $5P$ parameters, or twenty five in the main setting. The shared nuisance is integrated analytically through a covariance that is diagonal plus rank one. Only observed calibration entries contribute to each row's likelihood.

The response sign of each sensor is fixed using its support correlation. The generator's lower bound on response magnitude does not guarantee that the sign is recovered from every finite support set. Fixing it reduces the posterior being sampled and omits uncertainty over alternative signs. For this reason, the reference is called sign conditioned HMC throughout the manuscript. It is a useful numerical comparison, rather than a certified Bayes optimal ceiling for the full task prior.

The implementation uses four chains per task, each initialized from a maximum a posteriori fit. Warmup first adapts step size toward an acceptance probability of 0.75, estimates a diagonal mass matrix over one hundred and fifty iterations, and then performs one hundred and fifty further adaptation iterations. Fifty retained draws per chain are thinned by a factor of four, with twelve leapfrog steps per trajectory.

Let $\theta^{(b)}$ denote a retained parameter draw. The predictive distribution includes the information in the query readings themselves:

$$p(y \mid x_S, \mathcal{D}) \approx \frac{\sum_b p(y, x_S \mid \theta^{(b)})}{\sum_b p(x_S \mid \theta^{(b)})}. \tag{29}$$

The denominator is evaluated by trapezoidal quadrature on three hundred and one points. This query weighting differs from simply averaging $p(y \mid x_S, \theta^{(b)})$ under the support posterior with equal weights. Query readings can make some deployment parameter draws more plausible than others.

Across the panels on which this reference was run, the recorded convergence diagnostic $\widehat{R}$ was below 1.1 for at least 99.9% of identifiable parameters. A check with known generator parameters reproduces the exact oracle predictive calculation within 2×10^{-3} nats. These checks assess the implemented calculation and sampling behavior under the recorded setup. They do not eliminate finite Monte Carlo error or the response sign conditioning.

B.3. Positive variance and Gaussian readout

Each corrected noise variance multiplies a positive anchor variance by

$$\exp\{4 \tanh(s_j/4)\}. \tag{30}$$

For finite network outputs this multiplier is positive and bounded between e^{-4} and e^{4}. The construction prevents a negative measurement variance. The proper prior then keeps the joint precision positive definite in exact arithmetic. Implementations still require numerically stable linear algebra, especially when fitted sensor responses are large or the support covariance is nearly singular.

For $d = 1 + K$, accumulating P dense rank one site contributions requires $O(Pd^2)$ arithmetic. Solving the final d dimensional Gaussian system requires $O(d^3)$ arithmetic with a conventional dense factorization. These expressions describe the readout after site construction. They omit support fitting, neural feature construction and any transformer attention, so they are not a total inference runtime claim. In the reported $K = 1$ setting the Gaussian state has two coordinates.

For cavity refinement, each site is removed from the sum before forming its update. Since the remaining precision is the prior plus positive semidefinite sites, it is positive definite in exact arithmetic. Updates are synchronous: every new site in a round uses cavities built from the previous round's sites. This avoids making the calculation depend on an arbitrary sensor update order. The original learned model nevertheless depends on the chosen nuisance coordinates, as the factor orientation audit makes clear.

B.4. Training comparison

The compact cavity model and the transformer are trained on the same synthetic task distribution and query mask rule, but their optimization recipes are not identical. The compact network draws from a finite task pool, whereas the transformer receives fresh tasks at every step. Their batch shapes, learning rates, clipping thresholds and step counts also differ. The primary compact versus transformer result therefore compares the specified complete training procedures. The transformer and transformer site model share a more closely matched backbone and recipe, making that comparison more informative about the effect of the output construction.

The implementation registers the original transformer's generic output head inside its backbone, even when the lifted site model uses a separate site head for prediction. Registered parameter counts therefore include a small head that is not used by the lifted prediction route. This detail explains why the parameter count increases by the full additional site head rather than by the difference between two replacement heads.

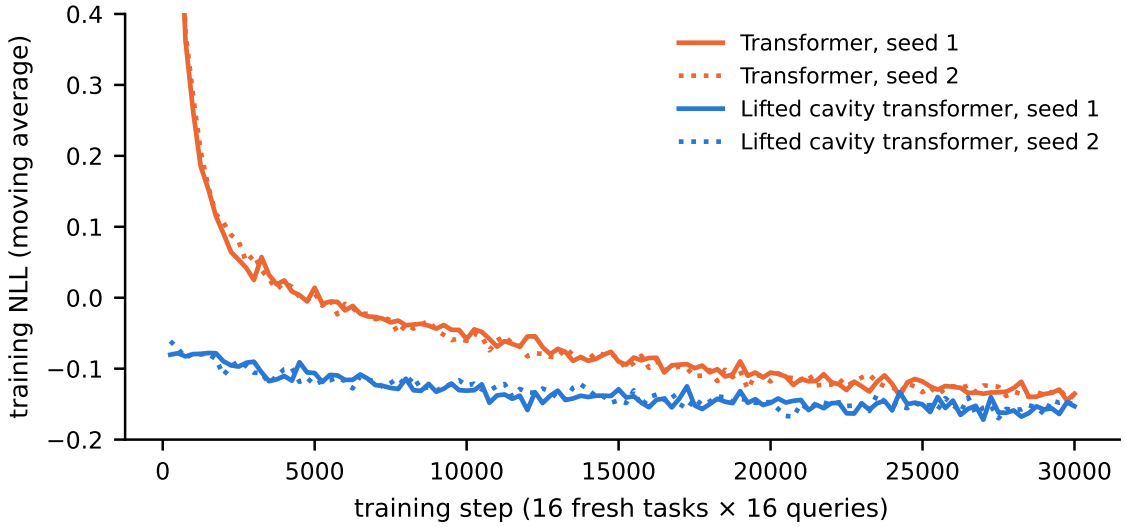


Figure 4: Historical training likelihood curves for the transformer and its lifted site variant. Both use the same backbone, recipe and training seed in the recorded comparison. The site output starts at a fitted Gaussian anchor. The conventional output must first learn a predictor from its generic initialization. These curves are retained from the original score based paper export.

Table 10
Protocol v1 endpoints (two missing sensors; positive gain favours the lifted cavity network; pass requires gain ≥ 0.01 and a positive lower bound).

Endpoint	Comparison	Gain [95% CI]	Result
E1	Fresh tasks, vs FA Gaussian	+0.074 [+0.064, +0.084]	pass
E2	Fresh tasks, vs residual MLP	+0.016 [+0.012, +0.020]	pass
E3	8 sensors, vs FA Gaussian	+0.115 [+0.097, +0.132]	pass
E4	Nonlinearity 0.8, vs residual MLP	+0.028 [+0.020, +0.036]	pass
R1	Air Quality NO_2, 23 later weeks, vs EM Gaussian	+0.112 [+0.050, +0.173]	pass

The architecture figures are conceptual renderings of these computations. Their support matrices and sensor traces are illustrative, and their analytical sign reversal example follows the stated Gaussian model. They do not display newly collected sensor data or newly measured benchmark outcomes.

C. Protocol history and execution audits

C.1. Protocol history and all declared outcomes

The development history is retained because it explains why the controls and hypotheses changed. An earlier three round scalar cavity network first collapsed to the prior. After repair with support moment initialization, its best seed scored 0.403 nats with two missing sensors, worse than the ridge control at 0.396; the oracle scored −0.046. Strengthening the support fitted references revealed scores of 0.168 for EM and 0.156 for factor analysis on that development panel. A larger training pool did not eliminate the subsequent scalar model deficit. These observations motivated lifting, but the later model's success does not erase the earlier failed configurations.

Protocol v1 was committed before its confirmation panels were generated and declared five endpoints. Every outcome is retained in Table 10: fresh task gains over FA Gaussian and the residual MLP, eight sensor transfer, stronger nonlinearity against the MLP, and Air Quality NO_2 against EM. All five recorded gates passed. The first endpoint additionally required every lifted training seed to gain at least 0.01 nats. The Air Quality comparison used 23 later weeks; it is one device and one fine tuning run for that original real endpoint. Git commit ordering and hashes recorded with results document the sequence, but do not independently prove that no earlier data access occurred.

Protocol v2 strengthened the synthetic and real controls and introduced the previously unscored Beijing $PM_{2.5}$ test period. Five of its six endpoints passed. E1 and E2 compare the small model with NL FA and the transformer on F1. E3a and E3b compare it with FA Gaussian and the transformer at sixteen sensors. E4a compares the adapted model with BLR on Beijing. E4b compares it with the adapted transformer and fails, with gain −0.118 [−0.145, −0.091]. A recorded change before any Beijing score supplied an empty feature fallback for BLR: it predicts the support mean and variance

Table 11
Every declared endpoint of protocols v2 to v4. Protocols v2/v3 preceded their panels; v4 was amended after partial data/reference inspection, before compared model scoring (protocol v1: Appendix C). Gain in nats of NLL, positive when the first named model is better; paired 95% intervals, computed across weeks on Beijing. v4 endpoints average three training seeds per task. Pass: gain ≥ 0.01 and lower bound > 0, except the noninferiority endpoints E7b and E10 (lower bound > -0.02).

ID	Data	Comparison	Gain [95% CI]	Result
E1	F1, $k = 2$	lifted cavity network vs NL FA	+0.054 [+0.043, +0.065]	pass
E2	F1, $k = 2$	lifted cavity network vs transformer	+0.033 [+0.027, +0.040]	pass
E3a	F3 (16 sensors), $k = 2$	lifted cavity network vs FA Gaussian	+0.148 [+0.115, +0.181]	pass
E3b	F3 (16 sensors), $k = 2$	lifted cavity network vs transformer	+0.144 [+0.127, +0.161]	pass
E4a	Beijing $PM_{2.5}$	lifted cavity network (FT) vs BLR	+0.194 [+0.131, +0.257]	pass
E4b	Beijing $PM_{2.5}$	lifted cavity network (FT) vs transformer (FT)	−0.118 [−0.145, −0.091]	**fail**
E5	G1, $k = 2$	LCT vs transformer	+0.035 [+0.027, +0.042]	pass
E6	G3 (16 sensors), $k = 2$	LCT vs transformer	+0.150 [+0.133, +0.167]	pass
E7a	Beijing NO_2	LCT (FT) vs lifted cavity network (FT)	+0.062 [+0.024, +0.100]	pass
E7b	Beijing NO_2	LCT (FT) vs transformer (FT), noninferiority	−0.021 [−0.093, +0.052]	**fail**
E8	H1, $k = 2$	LCT large vs transformer large	pending	n/a
E9	H3 (16 sensors), $k = 2$	LCT large vs transformer large	pending	n/a
E10	Beijing new	LCT large (FT) vs transformer large (FT), noninferiority	pending	n/a
E11	Beijing new, +6 removed	LCT large (FT) vs transformer large (FT)	pending	n/a
E12	Beijing new	LCT large (FT) vs TabPFN v2 (in context)	pending	n/a

$s^2(1 + 1/n)$ when no station is observed. This prevented a build failure without changing episodes or other methods.

Protocol v3 followed the real data loss and introduced the lifted transformer readout on new panels and targets. E5 and E6 pass on the five and sixteen sensor synthetic panels. E7a passes against the small adapted network on Beijing NO_2. E7b fails to establish the fixed noninferiority margin against the adapted transformer. Beijing CO was declared descriptive. LCT results on F1 and $PM_{2.5}$, additional training seeds, Gaussian processes, nearest neighbours and longer adaptation runs remain post hoc. Training interruptions and restarts were documented before v3 scoring. Checkpoint resume tests establish the tested CPU interruption behavior; they do not certify every subsequent GPU optimization or a full independent training rerun.

C.2. Factor orientation and historical reproducibility

An import audit provides a useful distinction between aggregate reproduction and exact prediction replay. It matched nine imported checkpoint hashes, regenerated the three v1 synthetic panels and the NO_2 test panel, and replayed the five endpoint decisions without fitting a new model. The largest discrepancy in endpoint interval limits was about 2.3×10^{-5} nats. However, the full cell replay gate failed: the eight sensor C2 panel differed by as much as 0.012804 nats. Reproduction of the pass decisions therefore cannot be reported as complete numerical reproduction of every saved prediction.

The diagnostic identified factor orientation as a cause. A loading vector D and its negative define the same residual covariance, but the original learned site network does not enforce equivalence of those representations. On C2 seed one, comparing both signs reduced the maximum replay discrepancy to less than 2.81×10^{-6} nats. Across tasks, changing sign could alter cell NLL by 0.08460. This explains a numerical vulnerability; selecting a sign using evaluation losses would change the inference rule and is not an acceptable correction to the historical scores. A permutation check that permutes an already fitted anchor likewise does not establish invariance after refitting the factor model.

A separate source gate evaluated a label independent wrapper that predicts under both signs and matches the first two moments. It passed six fixed deterioration checks and made predictions invariant to the sign of one factor. That is an engineering result for a separately identified version, not new real data efficacy evidence. It takes two forward passes, returns a Gaussian rather than the exact mixture density, and does not resolve arbitrary rotations or degenerate eigenspaces for multiple factors. Historical tables retain the original model and scores. The theoretical structural statements must therefore be read with their stated anchor orientation assumptions, rather than as an unconditional reproducibility guarantee for the imported implementation.

C.3. The v4 audit and withheld endpoints

The larger continuation fixed two 2.1 million parameter backbones, three training seeds, 40,000 synthetic updates, new H1 and H3 panels, and Beijing PM_{10}, SO_2 and O_3. It also specified pretrained TabPFN v2 for inference in context with eight estimators and a fixed random state. Before compared model scoring, a recorded amendment added uniform dequantization before taking logarithms [32]. The reason was discovered during panel construction: 34.2% of SO_2 and 12.8% of O_3 test readings were at the floor of 2 $\mu g\,m^{-3}$, and three SO_2 support sets were constant. Integer readings received fixed noise in $(-1/2, 1/2)$, shared by the constructed source and test episodes. The only prior outputs were closed form references from the discarded undequantized panels. This amendment preceded model comparisons but followed partial data inspection, which limits any claim of an entirely untouched protocol.

The bounded L40S continuation completed, and 24 learned checkpoint files and 35 expected score archives were collected. The strict verifier nevertheless stopped before endpoint arithmetic because twelve fine tuning runs from seeds two and three lacked the required checkpoint identity sidecar. The executed orchestration checked completion before launching a fresh fit, but omitted the completion and binding check after that subprocess returned. It could therefore proceed to scoring without recording the required checkpoint identity first. Six seed one sidecars existed in the final collection but were absent from its initial recovery snapshot, so their later presence does not establish creation before scoring.

All five v4 endpoints, E8 through E12, are withheld. Entries marked pending in the preserved endpoint table mean unverified and undisclosed, not an efficacy pass, failure or a claim that training is still running. CPU prediction replay, endpoint integration and secondary analyses were not executed after the gate failed. Current file hashes establish present byte identities; they cannot supply the missing historical attestation. The development driver now binds a completed fresh fit before scoring and refuses existing unbound scores, with four orchestration regression tests recorded as passing. This prospective repair does not repair the executed scientific record. A forensic replay could answer whether current checkpoints reproduce current score archives, but it could not retrospectively satisfy the failed provenance rule.

Even an eligible future version of E12 would compare a model adapted on target source data with TabPFN used only in context. Matching the episode support would not match prior target specific exposure, so that endpoint could not isolate architecture alone. The present withheld study supplies no evidence of better pretrained system performance.

D. Additional historical score tables

The tables below retain the remaining deletion conditions for the v2 width, response and support size panels, together with all reported Air Quality deletion conditions. They supplement the primary comparisons rather than introducing new endpoints. Transformer readout rows on these previously used panels remain descriptive, and device target rows retain their original confirmation or post hoc status.

Table 12
Panel F2 (8 sensors; 128 tasks): mean test NLL by number of missing sensors.

Method	k=0	k=1	k=2	k=3
Oracle	−0.807	−0.712	−0.598	−0.459
FA Gaussian	−0.462	−0.387	−0.296	−0.183
EM Gaussian	0.032	−0.133	−0.160	−0.120
NL FA	−0.485	−0.414	−0.327	−0.215
BLR	0.187	0.398	0.615	0.028
Ridge (complete case)	0.064	−0.003	−0.033	−0.015
Lifted cavity	−0.615	−0.530	−0.427	−0.302
Lifted cavity, $K = 2$	−0.618	−0.531	−0.427	−0.302
Lifted sites, no cavity	−0.551	−0.469	−0.369	−0.250
Scalar sites ($K = 0$)	−0.295	−0.228	−0.148	−0.057
Transformer	−0.568	−0.474	−0.362	−0.229
Transformer, all masks	−0.545	−0.452	−0.343	−0.212
Lifted cavity transformer (v3)	−0.611	−0.526	−0.423	−0.299

Table 13
Panel F3 (16 sensors; 128 tasks): mean test NLL by number of missing sensors.

Method	k=0	k=1	k=2	k=3
Oracle	−1.223	−1.184	−1.142	−1.096
FA Gaussian	−0.864	−0.832	−0.797	−0.758
EM Gaussian	6.300	4.851	2.977	1.553
NL FA	−0.910	−0.876	−0.841	−0.803
BLR	0.095	0.078	0.058	0.054
Ridge (complete case)	0.093	0.089	0.081	0.086
Lifted cavity	−1.018	−0.982	−0.945	−0.902
Lifted cavity, $K = 2$	−1.017	−0.981	−0.944	−0.901
Lifted sites, no cavity	−0.952	−0.916	−0.880	−0.836
Scalar sites ($K = 0$)	−0.658	−0.627	−0.591	−0.554
Transformer	−0.886	−0.846	−0.801	−0.747
Transformer, all masks	−0.869	−0.832	−0.790	−0.738
Lifted cavity transformer (v3)	−1.020	−0.985	−0.949	−0.905

Table 14
Panel F4 (nonlinearity 0.8; 128 tasks): mean test NLL by number of missing sensors.

Method	k=0	k=1	k=2	k=3
Oracle	−0.513	−0.331	−0.086	0.261
FA Gaussian	0.050	0.170	0.334	0.568
EM Gaussian	0.050	0.145	0.311	0.559
NL FA	−0.032	0.089	0.258	0.507
BLR	0.208	0.188	0.331	0.568
Ridge (complete case)	0.138	0.208	0.338	0.571
Lifted cavity	−0.185	−0.026	0.191	0.498
Lifted cavity, $K = 2$	−0.200	−0.035	0.186	0.497
Lifted sites, no cavity	−0.068	0.075	0.261	0.519
Scalar sites ($K = 0$)	−0.013	0.130	0.314	0.570
Transformer	−0.226	−0.043	0.191	0.503
Transformer, all masks	−0.221	−0.038	0.190	0.486
Residual MLP	−0.107	0.039	0.235	0.508
Scalar cavity network	0.133	0.256	0.416	0.635
Lifted cavity transformer (v3)	−0.196	−0.037	0.179	0.482

Table 15
Panel F5 (linear sensors; 128 tasks): mean test NLL by number of missing sensors.

Method	k=0	k=1	k=2	k=3
Oracle	−0.455	−0.288	−0.066	0.244
FA Gaussian	−0.362	−0.203	0.009	0.305
EM Gaussian	−0.244	−0.142	0.034	0.308
NL FA	−0.337	−0.178	0.030	0.318
BLR	−0.135	−0.085	0.061	0.322
Ridge (complete case)	−0.169	−0.092	0.062	0.323
Lifted cavity	−0.323	−0.172	0.031	0.318
Lifted cavity, $K = 2$	−0.320	−0.170	0.033	0.318
Lifted sites, no cavity	−0.343	−0.189	0.018	0.309
Scalar sites ($K = 0$)	−0.085	0.031	0.186	0.415
Transformer	−0.278	−0.122	0.085	0.371
Transformer, all masks	−0.281	−0.123	0.080	0.353
Residual MLP	−0.304	−0.161	0.036	0.319
Scalar cavity network	0.052	0.161	0.302	0.505
Lifted cavity transformer (v3)	−0.319	−0.171	0.029	0.315

Table 16
Panel F6 (16 support rows; 128 tasks): mean test NLL by number of missing sensors.

Method	k=0	k=1	k=2	k=3
Oracle	−0.446	−0.280	−0.060	0.252
FA Gaussian	0.256	0.371	0.510	0.686
EM Gaussian	24.900	4.682	1.140	0.739
NL FA	0.314	0.388	0.496	0.661
BLR	0.898	0.857	3.690	0.887
Ridge (complete case)	0.317	0.391	0.500	0.624
Lifted cavity	0.073	0.195	0.363	0.604
Lifted cavity, $K = 2$	0.156	0.250	0.392	0.614
Lifted sites, no cavity	0.117	0.234	0.389	0.604
Scalar sites ($K = 0$)	0.204	0.293	0.428	0.644
Transformer	0.253	0.371	0.523	0.731
Transformer, all masks	0.256	0.359	0.483	0.653
Residual MLP	0.094	0.211	0.371	0.594
Scalar cavity network	0.345	0.439	0.565	0.750
Lifted cavity transformer (v3)	0.237	0.333	0.457	0.643

Table 17
Panel F7 (96 support rows; 128 tasks): mean test NLL by number of missing sensors.

Method	k=0	k=1	k=2	k=3
Oracle	−0.429	−0.264	−0.043	0.269
FA Gaussian	−0.232	−0.082	0.114	0.385
EM Gaussian	−0.237	−0.086	0.112	0.385
NL FA	−0.276	−0.127	0.073	0.353
BLR	−0.178	−0.056	0.125	0.390
Ridge (complete case)	−0.163	−0.047	0.129	0.391
Lifted cavity	−0.324	−0.163	0.052	0.353
Lifted cavity, $K = 2$	−0.320	−0.159	0.056	0.357
Lifted sites, no cavity	−0.275	−0.123	0.078	0.359
Scalar sites ($K = 0$)	−0.097	0.034	0.205	0.443
Transformer	−0.295	−0.133	0.081	0.379
Transformer, all masks	−0.297	−0.134	0.080	0.367
Residual MLP	−0.287	−0.131	0.074	0.360
Scalar cavity network	0.047	0.161	0.309	0.516
Lifted cavity transformer (v3)	−0.329	−0.167	0.047	0.346

Table 18
Air Quality CO: mean test NLL for every number of simulated missing sensors (23 later weeks).

Method	k=0	k=1	k=2	k=3
EM Gaussian	0.074	0.063	0.085	0.146
FA Gaussian	0.052	0.086	0.116	0.154
NL FA	0.144	0.120	0.091	0.124
Bayesian linear regression	1.037	0.433	0.216	0.202
Ridge (complete case)	0.824	0.337	0.208	0.210
GP, complete case[†]	3.364	1.056	0.503	0.146
k nearest neighbours[†]	0.571	0.391	0.353	0.274
Lifted cavity, before adaptation	0.044	0.064	0.129	0.160
Transformer, before adaptation	−0.017	−0.006	0.081	0.154
Lifted cavity, fine tuned	−0.040	−0.031	0.008	0.077
no synthetic pretraining	−0.001	−0.014	−0.006	0.077
no cavity input	−0.086	−0.065	−0.028	0.038
scalar sites ($K = 0$)	−0.026	−0.021	−0.003	0.075
Transformer, fine tuned	−0.163	−0.149	−0.052	0.036
Residual MLP, fine tuned	0.025	0.061	0.110	0.160
Lifted cavity transformer, fine tuned[‡]	−0.099	−0.054	−0.027	0.044

Table 19
Air Quality NO_2: mean test NLL for every number of simulated missing sensors (23 later weeks).

Method	k=0	k=1	k=2	k=3
EM Gaussian	−0.237	−0.174	−0.175	−0.168
FA Gaussian	−0.196	−0.173	−0.155	−0.162
NL FA	−0.090	−0.102	−0.142	−0.169
Bayesian linear regression	−0.055	−0.214	−0.178	−0.160
Ridge (complete case)	−0.200	−0.230	−0.176	−0.145
GP, complete case[†]	25.217	0.696	−0.198	−0.258
k nearest neighbours[†]	−0.010	−0.020	−0.063	−0.128
Lifted cavity, before adaptation	−0.350	−0.295	−0.233	−0.176
Transformer, before adaptation	−0.360	−0.310	−0.231	−0.136
Lifted cavity, fine tuned	−0.412	−0.366	−0.337	−0.273
no synthetic pretraining	−0.419	−0.365	−0.316	−0.248
no cavity input	−0.423	−0.375	−0.332	−0.275
scalar sites ($K = 0$)	−0.359	−0.337	−0.302	−0.231
Transformer, fine tuned	−0.485	−0.429	−0.381	−0.294
Residual MLP, fine tuned	−0.246	−0.198	−0.147	−0.106
Lifted cavity transformer, fine tuned[‡]	−0.404	−0.396	−0.359	−0.292